RESIDUES OF A TROPICAL ZETA FUNCTION FOR CONVEX DOMAINS

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Abstract. For a compact convex domain $\Omega \subset \mathbb{R}^n$, we define an $SL(n,\mathbb{Z})$-invariant zeta function

$$Z_\Omega(s) := \int_\Omega \rho_\Omega(x)^{\,s-n}\, dV,$$

where ρ_Ω is the distance-to-the-boundary function determined by primitive lattice supporting lines.

In dimension 2, we introduce two further incarnations of this zeta function. First, we show that the integral zeta function can be rewritten, up to an explicit holomorphic correction term, as a tropical boundary zeta series indexed by Farey neighbors. Second, we show that this boundary series coincides with the zeta function obtained by summing the s-the powers of the coefficients of the Legendre dual in Hata's Schauder basis.

For polygons Ω with rational slope sides, the first pole of the tropical zeta function occurs at $s = 1$, and its residue is the lattice length of $\partial\Omega$.

For convex planar domains with C^3-smooth boundary and everywhere nonvanishing curvature, the $s = 1$ contribution disappears. We prove that $Z_\Omega(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, with a simple pole there, and

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{3^{5/2}}{2^{5/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\partial\Omega).$$

Hence the leading singularity in the smooth strictly convex case is governed by equiaffine boundary geometry.

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1. Introduction

1.1. **Broader view.** Gauss' circle problem [6, 3] is an archetypal problem at the interface of analytic number theory and planar geometry. Although it has been studied extensively for more than two hundred years, it still remains unsolved, despite substantial progress achieved through a broad range of methods and ideas. One of the aims of the present article is to introduce another tool into this circle of questions, stemming from recent developments in what may be called "non-algebraic tropical geometry" or "tropical optics". Namely, to a compact convex domain one can canonically associate a function of one complex variable, which we call its tropical zeta function. Just as the zeros of the Riemann zeta function control fine asymptotic properties of the distribution of prime numbers [20, 17], we expect the poles of the tropical zeta function to be crucial for obtaining sharp bounds on the error term in lattice-point counting for dilates of the domain.

Indeed, for a planar convex domain, the main term in lattice-point counting is quadratic and equals the area of the domain, whereas the general bound for the error term is linear in the dilation parameter R. The coefficient of this linear term is governed by the straight segments of the boundary, and therefore, in the strictly convex case, one can do better. If one further assumes that the boundary is sufficiently smooth and has everywhere nonvanishing curvature, then van der Corput proved [5] about a century ago that the error term is at most of order $R^{2/3}$, see also [11, 10]. In terms of tropical zeta functions, the first bound corresponds to the existence of a rightmost

pole at $s = 1$ for domains whose boundary contains rational straight segments, the residue being the lattice perimeter. For domains with sufficiently smooth boundary, the next pole, moving from right to left, appears at $s = \frac{2}{3}$, and the corresponding residue is universally proportional to the affine arc length of the boundary. This is the main and technically most difficult result of the paper.

The definition of the tropical zeta functions considered in this paper originates in the mathematical study of self-organized criticality. More precisely, finite-point perturbations of the maximal stable state in the Abelian sandpile model give rise to tropical analytic curves and to the tropical power series defining them [14, 13, 12, 15]. One such series, corresponding to the action of a point perturbation at the special central point, is the tropical distance function of a convex domain, while the associated tropical analytic curve is the so-called tropical caustic [18, 22]. The convex domain itself appears as the domain of convergence of this tropical series, which defines a positive concave piece-wise linear function on its interior. Integrating its s-th power over the domain then produces a function of the complex variable s, which we call the tropical zeta function of the domain. The central conjecture concerning this class of functions is that they admit meromorphic continuations to the entire complex plane.

One case in which the conjecture holds almost trivially is that of polygons whose sides have rational slopes. Such polygons are naturally viewed as moment polytopes of compact symplectic toric surfaces. From the algebro-geometric point of view, these surfaces may be obtained from a minimal model by a sequence of blow-ups at torus-fixed points. From the symplectic point of view, however, each blow-up comes with a natural "size", measured by the symplectic area of the corresponding exceptional divisor. In this setting, our zeta function may be interpreted as a Dirichlet generating series for the sizes of these blow-ups, with the symplectic structure encoded through tropical optics and with a particular blow-up chain chosen canonically. From this perspective, one sees that the corresponding zeta function extends meromorphically to the whole complex plane. Moreover, it has only two poles, both reflecting basic geometric information about the toric surface: the residue at $s = 1$ again corresponds to the symplectic area of the anticanonical class, while the residue at $s = 0$ gives its self-intersection please double check!.

Apart from this "trivial" algebraic case, there is essentially only one domain for which we can currently write the tropical zeta function explicitly in terms of known special functions. This domain arises as the limit shape obtained from the concentration of measure for the uniform distribution on lattice polygons contained in a fixed square, as the lattice mesh tends to zero [23, 2, 1]. For this particular domain, the tropical zeta function is given by

$$\left(8 - 2^{2-s}\frac{\zeta_{\mathrm{SU}(3)}(s)}{\zeta(3s)}\right)\frac{1}{s(s-1)},$$

(fractional self-intersection of "canonical class"?!) where $\zeta_{\mathrm{SU}(3)}$ denotes Witten's $SU(3)$ zeta function, arising in the context of two-dimensional Yang–Mills theory, and ζ is the Riemann zeta function. The pole structure of

$\zeta_{\mathrm{SU}(3)}$ was described by Romik [21], who showed that all of its poles are simple and occur at $s = \frac{2}{3}$ and at $s = \frac{1}{2} - k$ for $k = 0, 1, 2, \ldots$. As a consequence, the basic corollary that the lattice perimeter of the tropical wave front at time $t \to 0^+$, that is, the lattice perimeter of the t-level set of the tropical distance function, is of order $t^{1/3}$, can be sharpened in this special case: the second term in the asymptotic expansion of the tropical wave-front lattice perimeter is proportional to $t^{1/2}$, and, assuming the Riemann hypothesis, the third term is of order at most $t^{1/6}$. In fact, we expect this behaviour to be universal for sufficiently smooth strictly convex domains.

We would like to emphasize that, although the main body of the present article is devoted to the two-dimensional case, the underlying construction is meaningful in arbitrary dimensions. One may also hope that the convexity assumption can eventually be weakened, or perhaps removed entirely. In such a framework, it is conceivable that tropical zeta functions could be attached to a broader class of arithmetico-geometric objects and would define a homomorphism from a suitable motivic ring to the field of meromorphic functions, somewhat in the spirit of the Hodge–Deligne polynomial in birational geometry and of Igusa zeta functions in arithmetic geometry. Broader continuation questions, as well as higher-dimensional analogues of the affine residue, remain open.

1.2. **Main results.** Let $\Omega \subset \mathbb{R}^2$ be a compact convex domain with nonempty interior, and let

$$h_\Omega(u) = \min_{x \in \Omega} \langle u, x \rangle, \qquad u \in \mathbb{R}^2,$$

be its lower support function. We define the associated tropical distance-to-the-boundary function by

$$\rho_\Omega(x) = \min_{u \in \mathbb{Z}^2_{\mathrm{prim}}} \big(\langle u, x \rangle - h_\Omega(u) \big), \qquad x \in \Omega,$$

where $\mathbb{Z}^2_{\mathrm{prim}}$ denotes the primitive lattice directions. This function is concave on Ω, vanishes on $\partial\Omega$, and is positive in the interior. The tropical zeta function of Ω is then defined by

$$Z_\Omega(s) = \int_\Omega \rho_\Omega(x)^{\,s-2}\, dV,$$

initially for $\Re(s)$ sufficiently large. In this way one associates to Ω an $SL(2, \mathbb{Z})$-invariant zeta function built from the canonical tropical distance function.

The function $Z_\Omega(s)$ admits a natural Mellin interpretation. If

$$\Omega_t := \{x \in \Omega : \rho_\Omega(x) \geq t\}$$

denotes the inward tropical wave front at time t, and if $L_\Omega(t)$ denotes its lattice length, then

$$Z_\Omega(s) = \int_0^{m_\Omega} t^{s-2} L_\Omega(t)\, dt, \qquad m_\Omega := \max_\Omega \rho_\Omega.$$

Thus the tropical zeta function is the Mellin transform of the lattice-length profile of the tropical wave fronts. In particular, the singularities of $Z_\Omega(s)$ are governed by the short-time asymptotics of the boundary evolution.

Our first result concerns the polygonal case. If Ω is a polygon whose sides have rational slopes, then $Z_\Omega(s)$ extends meromorphically to the whole complex plane, and its first pole occurs at $s = 1$. The residue at this pole is the lattice length of $\partial\Omega$. Thus, for rational polygons, the leading singularity records the lattice-visible part of the boundary.

Our main result concerns the opposite regime, namely smooth strictly convex domains. Let $\Omega \subset \mathbb{R}^2$ be a compact convex domain with C^3-smooth boundary and everywhere nonvanishing curvature. Then the pole at $s = 1$ disappears, and the leading singularity moves to $s = \frac{2}{3}$. More precisely, we prove that $Z_\Omega(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, with a simple pole there, and

$$\operatorname{Res}_{s=2/3} Z_\Omega(s) = \frac{3^{5/2}}{2^{5/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \operatorname{Length}_{\text{equiaffine}}(\partial\Omega).$$

Equivalently,

$$\operatorname{Res}_{s=2/3} Z_\Omega(s) = \frac{3^{5/2}}{2^{5/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \int_{\partial\Omega} \kappa^{1/3}\, ds.$$

Thus, in the smooth strictly convex case, the leading singularity of the tropical zeta function is governed by equiaffine geometry.

The proof is based on a reduction from the interior Mellin integral to a boundary Dirichlet series. In dimension two, one associates to $\partial\Omega$ a boundary zeta series

$$F_{\partial\Omega}(s) = \sum_{\substack{a,b,c,d\in\mathbb{Z}\\ ad-bc=1}} f_{\partial\Omega}(a,b,c,d)^s,$$

where the summands are determined by neighboring primitive directions and the corresponding support defects. If $\widehat{\Omega}$ denotes the minimal model of Ω, then one has an exact identity

$$s(s-1)Z_\Omega(s) = -F_{\partial\Omega}(s) + H_{\widehat{\Omega}}(s),$$

where $H_{\widehat{\Omega}}(s)$ is an explicitly computable term coming from the minimal model. Near $s = \frac{2}{3}$, this term is holomorphic, so the singular behavior of $Z_\Omega(s)$ is entirely governed by the boundary series.

A model case is provided by the parabola. For the parabolic domain, the boundary zeta series can be computed explicitly and turns out to be the primitive Mordell–Tornheim series, equivalently Witten's $SU(3)$ zeta function after the natural normalization. This model already contains the pole at $s = \frac{2}{3}$ and explains the appearance of the same transcendental constant as in the general residue formula.

The parabolic case is the simplest model because the second derivative of the Legendre dual is constant; consequently, the support defect is governed only by the universal arithmetic weight attached to a Farey pair, and the boundary zeta series reduces to a primitive Mordell–Tornheim series. For a general smooth strictly convex arc, by contrast, the second derivative varies along the slope parameter, so one is led to a Farey sum with a nonconstant weight. The main analytic task is therefore to separate this slowly varying geometric factor from the arithmetic summation. We do this by replacing the exact defect by an endpoint model and then studying the resulting

weighted Farey series using Fejér approximation together with power-saving estimates for incomplete Kloosterman sums. This yields the residue as an averaged slope-space density

$$\big((g^*)''(\alpha)\big)^{2/3}\,d\alpha,$$

and Legendre duality converts this density into the equiaffine length element

$$\big((g^*)''(\alpha)\big)^{2/3}\,d\alpha = (g''(x))^{1/3}\,dx = \kappa^{1/3}\,ds.$$

Thus the explicit parabolic computation extends to the general case after decoupling the variable curvature from the Farey arithmetic.

Among zeta-type constructions attached to convex domains, the present one is especially well adapted to arithmetic questions: it is $SL(n,\mathbb{Z})$-invariant, so it respects the natural symmetry of the lattice, and in the planar smooth strictly convex case its leading nontrivial residue is the equiaffine length of the boundary. This is exactly the type of affine invariant that is known to control the behavior of rational and lattice points near convex curves; see [19, 9, 8]. In this sense, our zeta function is tailored to capture the arithmetic geometry of the boundary rather than only its Euclidean shape.

The paper is organized as follows. In Section 2, we begin by defining the tropical zeta function in integral form and deriving its Mellin representation. We then, Section 3, specialize to dimension two, introduce the boundary zeta series, and prove the integral–boundary identity. Routine computations as well as background material about the minimal model can be found in Appendix A. Toric/symplectic background and lattice perimeter asymptotics are in Appendix B. Next, in Section 4, we analyze the resulting Farey-type Dirichlet series and compute its residue at $s = \frac{2}{3}$. We then identify this residue with equiaffine length via Legendre duality. Heavy analytic details appear separately, in Appendix C. Finally, in Section 5, discuss the parabolic model and its relation to Mordell–Tornheim series and Witten's $SU(3)$ zeta function.

2. The tropical zeta function in the integral form

In this section we define the zeta function of a convex body by an integral over the body itself. In dimension 2, this integral presentation is equivalent, up to an explicit holomorphic correction term, to a boundary Dirichlet series built from primitive lattice directions and Farey neighbors.

2.1. **The tropical distance function.** Let $\Omega \subset \mathbb{R}^n$ be a compact convex body with nonempty interior. For

$$u \in \mathbb{R}^n$$

let

$$h_\Omega(u) := \min_{x\in\Omega}\langle u, x\rangle$$

be the lower support function of Ω. Denote by

$$\mathbb{Z}^n_{\mathrm{prim}} := \{u \in \mathbb{Z}^n \setminus \{0\} : \gcd(u_1,\dots,u_n) = 1\}$$

the set of primitive lattice directions.

Since

$$\langle mu, x\rangle - h_\Omega(mu) = m\big(\langle u, x\rangle - h_\Omega(u)\big) \qquad (m \in \mathbb{N}),$$

it is enough to minimize over primitive lattice vectors.

Definition 1. *For $x \in \Omega^\circ$, define the* tropical distance function *of x by*

$$\rho_\Omega(x) := \min_{u \in \mathbb{Z}^n_{\text{prim}}} \big(\langle u, x\rangle - h_\Omega(u)\big).$$

Indeed, if $x \in \Omega^\circ$, then

$$\langle u, x\rangle - h_\Omega(u) \geq c_x \|u\|$$

for some $c_x > 0$, so the minimum is attained.

The function ρ_Ω is continuous on Ω°, positive on the interior, and can be continuously extended to zero on $\partial\Omega$. It may be viewed as an $SL(n, \mathbb{Z})$-invariant analogue of a distance-to-the-boundary function, built from primitive lattice supporting hyperplanes rather than Euclidean normals.

Definition 2. *The* tropical zeta function (in the integral form) *of Ω is*

$$Z_\Omega(s) := \int_\Omega \rho_\Omega(x)^{\,s-n}\, dx,$$

initially defined for $\Re(s)$ sufficiently large.

The function inherits $\mathrm{SL}_n(\mathbb{Z})$- and translational- invariance, i.e.

$$Z_{A(\Omega)+v}(s) = Z_\Omega(s)$$

for all $A \in \mathrm{SL}_n(\mathbb{Z})$ and $v \in \mathbb{R}^n$, from the coordinate-free definition of ρ_Ω. As a conceptual remark, we would like to emphasize that the tropical zeta function alone is not enough to write an exact formula for the number of lattice points in the domain, as this number varies when a continuous family of translations is applied to the domain.

The normalization $s-n$ is chosen so that $Z_\Omega(n) = \mathrm{vol}(\Omega)$. More generally, this normalization assures that Z_Ω is homogeneous with respect to dilation of Ω in the following sense: $Z_{r\Omega}(s) = r^s Z_\Omega(s)$ for any positive r. In particular, a residue at some real point s_0 is a homogeneous invariant of Ω of degree s_0.

2.2. **Mellin form.** Let

$$M_\Omega := \max_{x \in \Omega} \rho_\Omega(x)$$

and, for $t \in [0, M_\Omega]$, define

$$\Omega_t := \{x \in \Omega : \rho_\Omega(x) \geq t\}.$$

Then for $\Re(s) > n$, the layer-cake formula gives

$$Z_\Omega(s) = (s-n) \int_0^{M_\Omega} t^{\,s-n-1}\, \mathrm{vol}(\Omega_t)\, dt.$$

Thus $Z_\Omega(s)$ is the Mellin transform of the volume profile of the family $\{\Omega_t\}_{t \geq 0}$.

For every $t > 0$, the set Ω_t is a convex polytope with finitely many facets, each of whose supporting hyperplanes has a primitive lattice normal.

Definition 3. *Let F be an $(n-1)$-dimensional polytope contained in an affine hyperplane*

$$H = \{x \in \mathbb{R}^n : \langle u, x\rangle = m\},$$

where $u \in \mathbb{Z}^n$ is primitive and $m \in \mathbb{Z}$. Its lattice volume *is defined by*

$$\mathrm{vol}_{\mathbb{Z}}(F) := \frac{\mathrm{vol}_{n-1}(F)}{\mathrm{covol}(H \cap \mathbb{Z}^n)},$$

where vol_{n-1} denotes Euclidean $(n-1)$-dimensional volume in H.

Definition 4. *For $t > 0$, the* lattice surface volume *of Ω_t is the sum of the lattice volumes of all facets of Ω_t. We denote it by $P_\Omega(t)$.*

Since $\mathrm{covol}(H \cap \mathbb{Z}^n)$ is equal to the length of the primitive integer normal vector to H, for $t > 0$, one has

$$P_\Omega(t) = -\frac{d}{dt}\,\mathrm{vol}(\Omega_t).$$

shall we prove it? it woluld be better to do so, in appendinx, along with a proof that of omega is a rational polytope then it it has the first pole and the rsidue is proportional to the lattice surface volume of it

Proposition 1 (Lattice surface volume form of the Mellin transform)**.** *For $\Re(s) > n$,*

$$Z_\Omega(s) = \int_0^{M_\Omega} t^{s-n} P_\Omega(t)\,dt.$$

Proof. Starting from

$$Z_\Omega(s) = (s-n)\int_0^{M_\Omega} t^{s-n-1}\,\mathrm{vol}(\Omega_t)\,dt,$$

integrate by parts:

$$(s-n)\int_0^{M_\Omega} t^{s-n-1}\,\mathrm{vol}(\Omega_t)\,dt = \Big[t^{s-n}\,\mathrm{vol}(\Omega_t)\Big]_0^{M_\Omega} - \int_0^{M_\Omega} t^{s-n}\frac{d}{dt}\,\mathrm{vol}(\Omega_t)\,dt.$$

Since $\mathrm{vol}(\Omega_{M_\Omega}) = 0$ and

$$t^{s-n}\,\mathrm{vol}(\Omega_t) \to 0 \qquad \text{as } t \to 0^+$$

for $\Re(s) > n$, this gives

$$Z_\Omega(s) = \int_0^{M_\Omega} t^{s-n}\left(-\frac{d}{dt}\,\mathrm{vol}(\Omega_t)\right)dt.$$

□

In particular, for $n = 2$, $Z_\Omega(s)$ is the Mellin transform of the lattice perimeter profile of the tropical wave fronts Ω_t.

Remark 1. It is natural to ask whether $Z_\Omega(s)$ satisfies a functional equation, or more generally whether its Mellin transform structure reflects a duality on tropical wave fronts. We do not know the answer.

3. The planar boundary zeta series

The following definition makes sense for arbitrary dimensional Ω.

Definition 5 (Minimal model)**.** *Let*

$$m_\Omega := \max_{x\in\Omega} \rho_\Omega(x), \qquad M_\Omega := \{x \in \Omega : \rho_\Omega(x) = m_\Omega\}.$$

Let $\mathcal{E}_\Omega$ be the set of all primitive vectors u for which there exist a point x in M_Ω satisfying $m_\Omega = \langle u, x\rangle - h_\Omega(u)$. The minimal model *of Ω is*

$$\hat{\Omega} = \bigcap_{u\in\mathcal{E}_\Omega} \{x \in \mathbb{R}^2 : \langle u, x\rangle \geq h_\Omega(u)\}.$$

Since $\mathcal{E}_\Omega$ is a finite set, $\hat{\Omega}$ is a polytope with rational slope sides. It is characterized by the properties that $m_\Omega = m_{\hat{\Omega}}$, $M_\Omega = M_{\hat{\Omega}}$, and $\rho_{\hat{\Omega}}$ coincides with ρ_Ω in a neighborhood of M_Ω. Clearly, $\Omega \subset \hat{\Omega}$.

Now we specialize to the two-dimensional case.

Definition 6. *A Γ-triangle is a shape bounded by an embedded planar arc Γ and two straight segments of rational slopes such that the primitive vectors in the directions of these segments form a basis of the lattice, and the segment connecting the endpoints of Γ together with Γ bound a convex region.*

From the discussion in appendix B.3, it follows that the complement of the interior of Ω in $\hat{\Omega}$ is a finite union of Γ-triangles (in fact, at most six of them due to Appendix A.2 where all types of minimal models are enumerated).

The basis condition (a.k.a unimodularity) implies that there is an $\mathrm{SL}_2(\mathbb{Z})$ transformation, which after being combined with a translation, puts the Γ-triangle in such a position that it belongs to the first quadrant and the straight segments of the triangle a parts of the coordinate axes. In particular, now Γ represents a graph of a convex function $g : [0, r] \mapsto [0, u]$ such that $g(r) = u$.

For each primitive vector $(a, b) \in \mathbb{Z}^2_{\geq 0} \setminus \{(0, 0)\}$, let

$$ax + by = \gamma_{a,b}$$

be the supporting line to the graph of g with normal (a, b). If

$$a, b, c, d \in \mathbb{Z}_{\geq 0}, \qquad ad - bc = 1,$$

define

$$f_\Gamma(a, b, c, d) := \left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|.$$

Definition 7. *The* boundary zeta series *attached to Γ is*

$$F_\Gamma(s) := \sum_{\substack{a,b,c,d\geq 0\\ ad-bc=1}} f_\Gamma(a, b, c, d)^s.$$

For the whole boundary (and assuming the adequate change of coordinates), we set

$$F_{\partial\Omega}(s) := \sum_\Gamma F_\Gamma(s),$$

where the sum runs over the several Γ-triangles forming $\hat{\Omega}\backslash\Omega^\circ$.

The integral zeta function and the boundary Dirichlet series are related by an exact identity.

Theorem 1. *Let $\Omega \subset \mathbb{R}^2$ be a compact convex domain, and let $\widehat{\Omega} \supset \Omega$ be its minimal model. Then for $\Re(s) > 2$,*

$$s(s-1)Z_\Omega(s) = -F_{\partial\Omega}(s) + H_{\widehat{\Omega}}(s),$$

where

$$H_{\widehat{\Omega}}(s) := s(s-1)Z_{\widehat{\Omega}}(s)$$

is a meromorphic function, explicitly computable from the lattice lengths of the strata of the minimal model $\widehat{\Omega}$.

See the proof on page 28, and tropical zeta functions of all minimal models are computed in A.2.

Remark 2. In the special case when $\widehat{\Omega} = [0, P] \times [0, Q]$ with $P \geq Q > 0$, one has four arcs and

$$H_{\widehat{\Omega}}(s) = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1}.$$

3.1. **The first singularity and equiaffine perimeter.** We now state the main boundary residue formula.

Theorem 2. *Let Γ be a C^3 convex planar arc with everywhere non-zero curvature. Then $F_\Gamma(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, with a simple pole at $s = \frac{2}{3}$, and*

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \frac{\sqrt{3}}{4^{1/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

See the proof on page 17.

Since $\partial\Omega$ is decomposed into three to six such arcs (accoring to the minimal model), summing Theorem 2 over these four pieces gives

$$\operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s) = \frac{\sqrt{3}}{4^{1/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\partial\Omega).$$

Combining this with the identity of Theorem 1, which for $\Re(s) > 2$ expresses $Z_\Omega(s)$ as

$$-\frac{F_{\partial\Omega}(s)}{s(s-1)}$$

plus a holomorphic term, we obtain a corresponding residue formula for the interior zeta function $Z_\Omega(s)$. In particular, $Z_\Omega(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, and

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{9}{2} \operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s).$$

Therefore

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{9\sqrt{3}}{2 \cdot 4^{1/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\partial\Omega).$$

Thus the first singularity of $Z_\Omega(s)$ is governed by the equiaffine perimeter of $\partial\Omega$.

Remark 3. We conjecture that, in the n-dimensional case, the first pole for the tropical zeta function of a strictly convex and sufficiently smooth domain occurs at

$$s = \frac{n(n-1)}{n+2},$$

and that the corresponding residue is proportional to the equiaffine surface volume of $\partial\Omega$. We are currently unable to prove this conjecture. In particular, the analytic continuation of the zeta function beyond this pole remains open.

We now recall the basic facts about equiaffine length used in the statement of the theorem.

3.2. **Equiaffine length.** Let $\Gamma : [0,1] \to \mathbb{R}^2$ be a C^2 regular curve,

$$\Gamma(t) = (x(t), y(t)), \qquad \dot{\Gamma}(t) \neq 0.$$

Its Euclidean curvature is

$$\kappa(t) = \frac{|\det(\dot{\Gamma}(t), \ddot{\Gamma}(t))|}{\|\dot{\Gamma}(t)\|^3},$$

and its equiaffine arclength element is

$$ds_{\text{equiaffine}} = |\det(\dot{\Gamma}(t), \ddot{\Gamma}(t))|^{1/3}\, dt.$$

Accordingly,

$$\text{Length}_{\text{equiaffine}}(\Gamma) = \int_\Gamma \kappa^{1/3}\, ds = \int_0^1 |\det(\dot{\Gamma}(t), \ddot{\Gamma}(t))|^{1/3}\, dt.$$

If Γ is written locally as the graph of a convex C^2-function $y = g(x)$, then

$$\text{Length}_{\text{equiaffine}}(\Gamma) = \int (g''(x))^{1/3}\, dx.$$

There is also a geometric interpretation in terms of support triangles: if Γ is partitioned into sufficiently small arcs and Δ_i denotes the support triangle corresponding to the i-th arc, then

$$\text{Length}_{\text{equiaffine}}(\Gamma) = \lim_{\max_i |t_i - t_{i-1}| \to 0} \sum_i \big(2\ \text{Area}(\Delta_i)\big)^{1/3}. \tag{1}$$

4. Farey intervals and the zeta function

Definition 8 (Farey interval)**.** *A* Farey interval *is an interval*

$$I = [c/d,\ a/b] \subset [0,1]$$

with $b, d \in \mathbb{N}$, $a, c \in \mathbb{Z}$ *satisfying* $0 \le c \le d$, $0 \le a \le b$, $\gcd(a,b) = \gcd(c,d) = 1$, *and*

$$ad - bc = 1.$$

We write its mediant as $m(I) = \frac{a+c}{b+d}$.

Definition 9 (Hata coefficient and its renormalization)**.** *Let $f \in C^2([0,1])$. For $I = [c/d, a/b]$ define*

$$c_I(f) := f\left(\frac{a+c}{b+d}\right) - \frac{b}{b+d} f\left(\frac{a}{b}\right) - \frac{d}{b+d} f\left(\frac{c}{d}\right). \tag{2}$$

Define

$$T_I(f) := (b+d)\, c_I(f).$$

Definition 10 (Farey zeta function)**.** *For $s \in \mathbb{C}$ with $\Re(s)$ large, define*

$$Z_f(s) := \sum_I |T_I(f)|^s,$$

where the sum runs over all Farey intervals $I \subset [0,1]$.

Remark 4 (On residues)**.** Throughout, $\mathrm{Res}_{s=2/3}$ denotes the residue of a *meromorphic continuation* of the relevant function to a punctured neighborhood of $s = \frac{2}{3}$. Accordingly, whenever we compute a residue we also prove that, after subtracting the explicit polar part, the remainder is holomorphic on some disk $|s - \frac{2}{3}| < \eta$.

Remark 5. Removing or adding finitely many intervals (e.g. those adjacent to 0/1 or 1/1) changes $Z_f(s)$ by an entire function. Hence it does not affect the residue at $s = \frac{2}{3}$.

Assume from now on that

$$f \in C^3([0,1]) \quad \text{and} \quad 0 < m \le |f''(x)| \le M \text{ for all } x \in [0,1]. \tag{3}$$

Theorem 3 (Residue formula)**.** *Assume* (3)*. Then $Z_f(s)$ admits a meromorphic continuation to a disk $|s - \frac{2}{3}| < \eta$ (for some $\eta > 0$) with a simple pole at $s = \frac{2}{3}$. Moreover,*

$$\mathrm{Res}_{s=2/3}\, Z_f(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3} \int_0^1 |f''(v)|^{2/3}\, dv.$$

See the proof on page 15. We first reduce the geometric quantity $T_I(f)$ to the local curvature f'' on the interval I. This leads to an endpoint model $Z_f^{\mathrm{end}}(s)$, which has the same residue at $s = \frac{2}{3}$. We then reparametrize Farey intervals by coprime denominator pairs, reorganize the series by the first denominator b, and show that the inner sums admit an equidistributed asymptotic with a power-saving error. The pole then comes entirely from the classical Dirichlet series $\sum_{b \ge 1} \varphi(b) b^{-3s}$.

To every Farey interval I, Hata associates [7] the piecewise-linear function

$$S_I(x) := \frac{b+d}{2}\Big(|a - bx| + |c - dx| - |a + c - (b+d)x|\Big).$$

Lemma 1. *For every Farey interval $I = \left[\frac{c}{d}, \frac{a}{b}\right]$, the function S_I satisfies*

$$S_I(x) \ge 0, \qquad \mathrm{supp}(S_I) = I, \qquad S_I\left(\frac{c}{d}\right) = S_I\left(\frac{a}{b}\right) = 0, \qquad S_I(m_I) = 1.$$

Moreover, S_I is piecewise linear and attains its unique maximum at the mediant m_I.

Theorem 4. *[7] Every continuous function $f \in C[0,1]$ admits the expansion*

$$f(x) = f(0) + (f(1) - f(0))x + \sum_I c_I(f)\, S_I(x),$$

where the sum is taken over all Farey intervals in Stern–Brocot order and converges uniformly on $[0,1]$. The coefficient of S_I is given by (2).

For our purposes, the importance of Hata's coefficient is that after the normalization $T_I(f) = (b+d)c_I(f)$, it behaves like a second-order local quantity attached to the interval I. Thus the zeta function $Z_f(s)$ is expected to be controlled by the local curvature f'', with a geometric weight depending only on the Farey denominators $b, d, b+d$. The next lemma makes this heuristic exact.

Lemma 2 (Mean-value formula for $T_I(f)$, proof on page A.4)**.** *Let $I = [c/d, a/b]$ be a Farey interval and $f \in C^2([0,1])$. Then there exists $\xi_I \in (c/d, a/b)$ such that*

$$T_I(f) = -\frac{f''(\xi_I)}{2\, bd(b+d)}.$$

Thus $c_I(f)$ is a weighted discrete second derivative of u along the Farey interval I. Lemma 2 shows that each summand of $Z_f(s)$ is governed by the value of f'' at some interior point ξ_I, multiplied by the universal arithmetic weight $(bd(b+d))^{-1}$. To isolate the arithmetic structure, we now replace the interior point ξ_I by one endpoint. This produces a simpler model series $Z_f^{\mathrm{end}}(s)$, and the next lemma shows that this modification does not change the residue at $s = \frac{2}{3}$.

Define

$$Z_f^{\mathrm{end}}(s) := 2^{-s} \sum_{I=[c/d,a/b]} \frac{|f''(a/b)|^s}{(bd(b+d))^s}.$$

Replacing ξ_I by an endpoint does not change the residue, as it is proved in the next lemma.

Lemma 3 (Proof on page 38)**.** *There exists $\eta_0 > 0$ such that $Z_f(s) - Z_f^{\mathrm{end}}(s)$ is holomorphic on the disk $|s - \frac{2}{3}| < \eta_0$. In particular, $\operatorname{Res}_{s=2/3} Z_f = \operatorname{Res}_{s=2/3} Z_f^{\mathrm{end}}$.*

Lemma 4 (Parametrization by modular inverses)**.** *Farey intervals $I = [c/d, a/b]$ (with $b, d \geq 1$) are in bijection with coprime pairs $(b,d) \in \mathbb{N}^2$ via*

$$a \equiv d^{-1} \pmod{b}, \quad 1 \leq a \leq b, \qquad c = \frac{ad-1}{b}.$$

Proof. If $ad - bc = 1$, then $ad \equiv 1 \pmod{b}$, so $(b,d) = 1$ and $a \equiv d^{-1} \pmod{b}$. Conversely, for coprime (b,d) choose a with $ad \equiv 1 \pmod{b}$ and define $c = (ad-1)/b \in \mathbb{Z}$; then $ad - bc = 1$. □

The parametrization by (b,d) separates the Farey combinatorics into an outer denominator b and an inner progression $d \equiv \overline{r} \pmod{b}$. After writing $d = kb + r$, the dependence on k becomes one-dimensional and is absorbed into a universal kernel $H_s(r/b)$. In this way the original Farey sum is reorganized into outer sums over b and inner reduced-residue sums over r.

For $\Re(s) > \frac{1}{2}$ and $u \in (0,1]$, define

$$H_s(u) := \sum_{k=0}^{\infty} (k+u)^{-s}(k+1+u)^{-s}.$$

For $\Re(s) > \frac{1}{2}$, using Lemma 4 and writing $d = kb + r$ with $(r,b) = 1$, one obtains

$$Z_f^{\mathrm{end}}(s) = 2^{-s} \sum_{b \geq 1} b^{-3s} \sum_{\substack{1 \leq r \leq b \\ (r,b)=1}} |f''(\overline{r}/b)|^s \, H_s(r/b), \tag{4}$$

where $\overline{r}$ denotes the inverse of r modulo b in $\{1, \ldots, b\}$.

Lemma 5 (Integral of H_s)**.** *For $\frac{1}{2} < \Re(s) < 1$,*

$$\int_0^1 H_s(u)\, du = \int_0^{\infty} t^{-s}(1+t)^{-s}\, dt = \frac{\Gamma(1-s)\Gamma(2s-1)}{\Gamma(s)}.$$

In particular, $\int_0^1 H_{2/3}(u)\, du = \Gamma(1/3)^2/\Gamma(2/3)$.

Proof. For $\Re(s) > \frac{1}{2}$, absolute convergence gives

$$\int_0^1 H_s(u)\, du = \sum_{k \geq 0} \int_0^1 (k+u)^{-s}(k+1+u)^{-s}\, du$$

by Fubini's theorem. Substitute $t = k + u$ in each term and note that the intervals $[k, k+1]$ partition $[0, \infty)$. This yields $\int_0^{\infty} t^{-s}(1+t)^{-s} dt$, which equals the Beta integral. □

The decomposition (4) reduces the residue problem to understanding, for each fixed b, the reduced-residue average of the product $H_s(r/b)\, |f''(\overline{r}/b)|^s$. We denote this inner sum by $\Sigma_b(s)$. The key analytic step is that $\Sigma_b(s)$ admits an asymptotic formula with a power-saving error, uniformly in s on compact subsets of $\{1/2 < \Re(s) < 1\}$.

Define

$$\Sigma_b(s) := \sum_{\substack{1 \leq r \leq b \\ (r,b)=1}} H_s(r/b)\, |f''(\overline{r}/b)|^s.$$

We prove the following equidistribution estimate for $\Sigma_b(s)$.

Proposition 2 (Proof on page 44)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$ and assume* (3)*. Then for every $\varepsilon > 0$ and $s \in K$ we have*

$$\Sigma_b(s) = \varphi(b) \left(\int_0^1 H_s(u)\, du \right) \left(\int_0^1 |f''(v)|^s\, dv \right) + O_{f,K,\varepsilon}\big(b^{\frac{1+\Re(s)}{2}+\varepsilon}\big),$$

with implied constant uniform for $s \in K$.

At this point the analytic difficulty is essentially finished. Indeed, Proposition 2 turns $Z_f^{\mathrm{end}}(s)$ into an explicit main term involving $\sum_{b \geq 1} \varphi(b) b^{-3s}$, plus an error which is holomorphic near $s = \frac{2}{3}$. Thus the pole and its residue come from the Euler-factor Dirichlet series for φ.

Proof of Theorem 3. By Lemma 3, it suffices to compute the residue of $Z_f^{\mathrm{end}}(s)$. Fix a disk $D := \{s : |s - 2/3| \le \eta\}$ with $0 < \eta < 1/30$ and set $\sigma_- := 2/3 - \eta$, $\sigma_+ := 2/3 + \eta$.

From (4) and Proposition 2 (applied on a compact set containing D but lying in $\{1/2 < \Re(s) < 1\}$), we get for $s \in D$:

$$Z_f^{\mathrm{end}}(s) = 2^{-s}\Big(\int_0^1 H_s\Big)\Big(\int_0^1 |f''(v)|^s\Big)\sum_{b\ge 1}\frac{\varphi(b)}{b^{3s}} + E(s),$$

where

$$E(s) = 2^{-s}\sum_{b\ge 1} b^{-3s}\, R_b(s), \qquad |R_b(s)| \ll_{f,D,\varepsilon} b^{\frac{1+\Re(s)}{2}+\varepsilon}.$$

In particular, for $s \in D$,

$$|b^{-3s}R_b(s)| \ll_{f,D,\varepsilon} b^{-3\sigma_- + \frac{1+\sigma_+}{2}+\varepsilon} = b^{-\frac{7}{6}+\frac{7\eta}{2}+\varepsilon}.$$

Choose $\varepsilon > 0$ so small that $-\frac{7}{6} + \frac{7\eta}{2} + \varepsilon < -1$ (possible since $\eta < 1/30$). Then $\sum_{b\ge 1} b^{-7/6+7\eta/2+\varepsilon}$ converges. Hence the series defining $E(s)$ converges uniformly on D by the Weierstrass M-test. Since each term is holomorphic in s, $E(s)$ is holomorphic on the interior of D.

The factor $A(s) := 2^{-s}\big(\int_0^1 H_s\big)\big(\int_0^1 |f''(v)|^s\big)$ is holomorphic on $\{1/2 < \Re(s) < 1\}$ by Lemma 5 and dominated convergence. The Dirichlet series identity $\sum_{n\ge 1}\varphi(n)n^{-w} = \zeta(w-1)/\zeta(w)$ valid for $\Re(w) > 2$ gives a meromorphic continuation of $\sum_{b\ge 1}\varphi(b)b^{-3s}$ near $s = 2/3$, with a simple pole at $3s = 2$ and residue $1/(3\zeta(2))$. Therefore $Z_f^{\mathrm{end}}(s)$ (hence $Z_f(s)$) is meromorphic on a neighborhood of $2/3$ with

$$\mathrm{Res}_{s=2/3}\, Z_f(s) = A(2/3)\cdot\frac{1}{3\zeta(2)}.$$

Now $A(2/3) = 2^{-2/3}\left(\int_0^1 H_{2/3}\right)\left(\int_0^1 |f''(v)|^{2/3}dv\right)$. By Lemma 5, $\int_0^1 H_{2/3} = \Gamma(1/3)^2/\Gamma(2/3)$. Using $\zeta(2) = \pi^2/6$ and $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$, the constant simplifies to $\sqrt{3}\,\Gamma(1/3)^3/(2^{2/3}\pi^3)$. □

4.1. **Hata's basis and Legendre duality.** We now explain that the boundary zeta series is, up to finitely many harmless terms, exactly the zeta series attached to the Hata coefficients of the Legendre dual. Thus Theorem 2 is a direct consequence of Theorem 3 applied to a suitable dual function.

Let $\Gamma \subset \partial\Omega$ be one of the four arcs cut out by the chosen horizontal and vertical supporting lines. After an $SL(2,\mathbb{Z})$ transformation followed by a translation, we may assume that Γ is the graph of a convex decreasing C^3-function

$$y = g(x), \qquad x \in [0,1],$$

and that the parameter

$$u = -g'(x)$$

runs through $[0,1]$ along Γ.

Define

$$\widetilde{g}(u) := g^*(-u), \qquad u \in [0,1],$$

where

$$g^*(\alpha) := \sup_{x\in\mathbb{R}}(\alpha x - g(x))$$

is the Legendre transform of g.

A supporting line to $y = g(x)$ with primitive integer normal $(a, b) \in \mathbb{Z}^2_{\geq 0} \setminus \{(0, 0)\}$ has equation

$$ax + by = \gamma_{a,b}.$$

If $b > 0$, then

$$y = -\frac{a}{b}x + \frac{\gamma_{a,b}}{b}.$$

Its slope is $-a/b$, and comparison with the Legendre representation

$$y = \alpha x - g^*(\alpha)$$

shows that

$$\widetilde{g}\Big(\frac{a}{b}\Big) = g^*\Big(-\frac{a}{b}\Big) = -\frac{\gamma_{a,b}}{b}.$$

Thus the values of $\widetilde{g}$ at rational points encode the support data of Ω.

Lemma 6 (Identification of the geometric weights)**.** *Let*

$$I = \Big[\frac{c}{d}, \frac{a}{b}\Big]$$

be a Farey interval, and let $v_I = (a + c, b + d)$ be the associated primitive vector. Then

$$T_I(\widetilde{g}) := (b + d)c_I(\widetilde{g}) = -\gamma_{a+c,b+d} + \gamma_{a,b} + \gamma_{c,d}.$$

In particular,

$$|T_I(\widetilde{g})| = f_\Gamma(a, b, c, d).$$

Proof. By the coefficient formula,

$$\begin{aligned} c_I(\widetilde{g}) &= \widetilde{g}\Big(\frac{a+c}{b+d}\Big) - \frac{b}{b+d}\widetilde{g}\Big(\frac{a}{b}\Big) - \frac{d}{b+d}\widetilde{g}\Big(\frac{c}{d}\Big) \\ &= -\frac{\gamma_{a+c,b+d}}{b+d} + \frac{\gamma_{a,b}}{b+d} + \frac{\gamma_{c,d}}{b+d}, \end{aligned}$$

and multiplying by $b + d$ gives the formula. □

The previous lemma shows that the geometric Dirichlet series attached to Γ is exactly the Hata-coefficient zeta series for the dual function $\widetilde{g}$, up to finitely many exceptional terms near the ends of the arc. As usual, such finitely many terms contribute an entire function and therefore do not affect residues.

Corollary 1. *There exists an entire function $E_\Gamma(s)$ such that*

$$F_\Gamma(s) = Z_{\widetilde{g}}(s) + E_\Gamma(s).$$

In particular,

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \operatorname*{Res}_{s=2/3} Z_{\widetilde{g}}(s).$$

It remains to identify the integral appearing in Theorem 3 with the equiaffine length of Γ.

Lemma 7 (Legendre duality and equiaffine length)**.** *The function $\widetilde{g}$ satisfies the assumptions of Theorem 3. Moreover,*

$$\int_0^1 (\widetilde{g}''(u))^{2/3}\,du = \int_0^1 (g''(x))^{1/3}\,dx = \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

Proof. Since $g \in C^3$ is convex and has a non-zero curvature on a compact interval, there exist constants $0 < m \le M < \infty$ such that

$$0 < m \le g''(x) \le M \qquad (x \in [0,1]).$$

Standard Legendre duality implies that $\widetilde{g} \in C^3([0,1])$ and, if $u = -g'(x)$, then

$$\widetilde{g}''(u) = \frac{1}{g''(x)}.$$

Also,

$$du = -g''(x)\,dx.$$

Since $u = -g'(x)$ decreases from 1 to 0 as x increases from 0 to 1, we have

$$\int_0^1 (\widetilde{g}''(u))^{2/3}\,du = \int_1^0 (g''(x))^{-2/3}\,(-g''(x))\,dx = \int_0^1 (g''(x))^{1/3}\,dx.$$

This proves the claim.

□

Proof of Theorem 2. By Corollary 1, the residue of $F_\Gamma(s)$ at $s = \frac{2}{3}$ is the same as the residue of $Z_{\widetilde{g}}(s)$. Applying Theorem 3 with $f = \widetilde{g}$, we obtain

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3} \int_0^1 (\widetilde{g}''(u))^{2/3}\,du.$$

By Lemma 7,

$$\int_0^1 (\widetilde{g}''(u))^{2/3}\,du = \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

Hence

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3}\,\mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

This proves the theorem. □

4.2. **Geometric meaning of the terms of the boundary zeta series.** The summands of the boundary zeta series admit a simple geometric interpretation in terms of small triangles cut out by supporting lines.

Let (a,b) and (c,d) be a unimodular pair, so that $ad - bc = 1$, and let $\Delta_{a,b,c,d}$ be the triangle bounded by the three supporting lines with primitive normals

$$(a,b), \qquad (c,d), \qquad (a+c, b+d).$$

Then

$$\mathrm{Area}(\Delta_{a,b,c,d}) = \frac{1}{2}\Big(\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\Big)^2.$$

Indeed, after applying an $SL(2,\mathbb{Z})$ transformation and then a translation, we may reduce to the case

$$(a,b) = (1,0), \qquad (c,d) = (0,1), \qquad \gamma_{a,b} = \gamma_{c,d} = 0.$$

In these coordinates, the three supporting lines are

$$x = 0, \qquad y = 0, \qquad x + y = \gamma_{a+c,b+d},$$

so $\Delta_{a,b,c,d}$ is a right isosceles triangle whose legs have length

$$\left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|.$$

Its area is therefore exactly

$$\frac{1}{2}\Big(\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\Big)^2.$$

Consequently, the terms in

$$F_\Gamma(s) = \sum_{\substack{a,b,c,d\geq 0\\ ad-bc=1}} f_\Gamma(a,b,c,d)^s$$

are given by

$$f_\Gamma(a,b,c,d) = \left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right| = \sqrt{2\ \mathrm{Area}(\Delta_{a,b,c,d})}.$$

Thus the coefficients of Hata's expansion of the Legendre dual have a direct geometric meaning: they measure the sizes of the support triangles determined by neighboring primitive normals.

Theorem 2 shows that the first singularity of the boundary Dirichlet series is governed by the asymptotic distribution of these small support triangles. Since

$$f_\Gamma(a,b,c,d) = \sqrt{2\ \mathrm{Area}(\Delta_{a,b,c,d})},$$

the residue at $s = \frac{2}{3}$ may be viewed as encoding a 1/3-power summation law for the areas of these triangles. This is precisely the exponent that appears in the formula (1) for the equiaffine length.

5. Example: The parabolic model

In this section we compute explicitly the boundary zeta series for the parabolic arc

$$\Gamma_{\mathrm{par}} := \{(x,y) \in \mathbb{R}^2_{\geq 0} :\ \sqrt{x} + \sqrt{y} = 1\},$$

and then deduce a closed formula for the tropical zeta function of the symmetric domain

$$L = \Big\{(x,y) \in \mathbb{R}^2 :\ \sqrt{1-|x|} + \sqrt{1-|y|} \geq 1\Big\}.$$

The resulting Dirichlet series is exactly the primitive Mordell–Tornheim series, hence is expressed in terms of Witten's $SU(3)$ zeta function.

5.1. **The parabolic arc.** Let Γ_{par} be the arc joining $(1,0)$ and $(0,1)$, and let

$$\Omega_{\mathrm{par}} := \{(x,y) \in \mathbb{R}^2_{\geq 0} :\ x + y \leq 1,\ \sqrt{x} + \sqrt{y} \geq 1\}$$

be the convex domain bounded by Γ_{par} and the two coordinate axes.

For a primitive vector $(a,b) \in \mathbb{Z}^2_{\geq 0} \setminus \{(0,0)\}$, let

$$ax + by = \gamma_{a,b}$$

be the supporting line to Ω_{par} with inward normal (a,b).

Lemma 8. *For every $a, b \geq 0$, not both zero, one has*

$$\gamma_{a,b} = \frac{ab}{a+b},$$

with the obvious convention that $\gamma_{1,0} = \gamma_{0,1} = 0$.

Proof. The boundary arc admits the parametrization

$$x = t^2, \qquad y = (1-t)^2, \qquad 0 \leq t \leq 1.$$

Hence

$$ax + by = at^2 + b(1-t)^2.$$

Differentiating with respect to t, we obtain

$$\frac{d}{dt}\big(at^2 + b(1-t)^2\big) = 2(a+b)t - 2b,$$

so the minimum is attained at

$$t = \frac{b}{a+b}.$$

Substituting this value gives

$$\gamma_{a,b} = a\Big(\frac{b}{a+b}\Big)^2 + b\Big(\frac{a}{a+b}\Big)^2 = \frac{ab}{a+b}.$$

□

Now let (a, b) and (c, d) be a unimodular pair, so that

$$a, b, c, d \in \mathbb{Z}_{\geq 0}, \qquad ad - bc = 1.$$

Recall that

$$f_{\Gamma_{\mathrm{par}}}(a, b, c, d) = \big|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\big|.$$

Proposition 3. *For every unimodular pair $(a, b), (c, d)$, one has*

$$f_{\Gamma_{\mathrm{par}}}(a, b, c, d) = \frac{1}{(a+b)(c+d)(a+b+c+d)}.$$

Proof. By Lemma 8,

$$\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d} = \frac{ab}{a+b} + \frac{cd}{c+d} - \frac{(a+c)(b+d)}{a+b+c+d}.$$

A direct simplification gives

$$\frac{ab}{a+b} + \frac{cd}{c+d} - \frac{(a+c)(b+d)}{a+b+c+d} = -\frac{(ad-bc)^2}{(a+b)(c+d)(a+b+c+d)}.$$

Since $ad - bc = 1$, taking the absolute value yields

$$f_{\Gamma_{\mathrm{par}}}(a, b, c, d) = \frac{1}{(a+b)(c+d)(a+b+c+d)}.$$

□

5.2. **The boundary series of the parabola.** We now identify the boundary zeta series of Γ_{par} with a primitive Mordell–Tornheim series.

Lemma 9. *The map*

$$(a,b,c,d) \longmapsto (p,q) := (a+b, c+d)$$

is a bijection between the set of quadruples

$$a,b,c,d \in \mathbb{Z}_{\geq 0}, \qquad ad - bc = 1,$$

and the set of coprime pairs $(p,q) \in \mathbb{N}^2$ *with* $\gcd(p,q) = 1$.

Proof. If $p = a+b$ and $q = c+d$, then

$$pd - qb = (a+b)d - (c+d)b = ad - bc = 1,$$

hence $\gcd(p,q) = 1$.

Conversely, let $p, q \in \mathbb{N}$ be coprime. Choose the unique integer b with

$$0 \leq b < p, \qquad qb \equiv -1 \pmod{p}.$$

Then

$$d := \frac{qb+1}{p} \in \mathbb{N}.$$

Since $0 \leq b < p$, we have $0 < d \leq q$. Now define

$$a := p - b, \qquad c := q - d.$$

Then $a, b, c, d \in \mathbb{Z}_{\geq 0}$, and

$$ad - bc = (p-b)d - b(q-d) = pd - bq = 1.$$

Thus every coprime pair (p,q) arises. Uniqueness follows from the uniqueness of b modulo p in the range $0 \leq b < p$. □

Proposition 4. *For* $\Re(s)$ *sufficiently large,*

$$F_{\Gamma_{\mathrm{par}}}(s) = \sum_{\substack{a,b,c,d \geq 0 \\ ad-bc=1}} \frac{1}{\big((a+b)(c+d)(a+b+c+d)\big)^s} = \sum_{\substack{p,q \geq 1 \\ (p,q)=1}} \frac{1}{\big(pq(p+q)\big)^s}.$$

Consequently,

$$F_{\Gamma_{\mathrm{par}}}(s) = \frac{1}{\zeta(3s)} \sum_{p,q \geq 1} \frac{1}{\big(pq(p+q)\big)^s}.$$

Proof. The first identity follows from Proposition 3 and Lemma 9.

For the second, write

$$\sum_{p,q \geq 1} \frac{1}{\big(pq(p+q)\big)^s} = \sum_{m \geq 1} \sum_{\substack{u,v \geq 1 \\ (u,v)=1}} \frac{1}{\big(mu \cdot mv \cdot m(u+v)\big)^s}.$$

This becomes

$$\sum_{m \geq 1} \frac{1}{m^{3s}} \sum_{\substack{u,v \geq 1 \\ (u,v)=1}} \frac{1}{\big(uv(u+v)\big)^s} = \zeta(3s)\, F_{\Gamma_{\mathrm{par}}}(s),$$

which proves the claim. □

It is therefore natural to introduce

$$\zeta_{\mathrm{SU}(3)}(s) := 2^s \sum_{p,q\geq 1} \frac{1}{\big(pq(p+q)\big)^s},$$

so that

$$F_{\Gamma_{\mathrm{par}}}(s) = 2^{-s}\frac{\zeta_{\mathrm{SU}(3)}(s)}{\zeta(3s)}.$$

5.3. **The special domain L.** We now return to the symmetric domain

$$L = \Big\{(x,y)\in\mathbb{R}^2 :\ \sqrt{1-|x|}+\sqrt{1-|y|}\geq 1\Big\}.$$

Its boundary consists of four congruent parabolic arcs. In the first quadrant the boundary is given by

$$\sqrt{1-x}+\sqrt{1-y}=1, \qquad 0\leq x\leq 1, \quad 0\leq y\leq 1.$$

Equivalently, after the affine change of variables

$$u = 1-x, \qquad v = 1-y,$$

this arc is identified with Γ_{par}.

Proposition 5. *The equiaffine arclength of one parabolic arc of ∂L is $4^{1/3}$. Hence*

$$\mathrm{Length}_{\mathrm{equiaffine}}(\partial L) = 4^{4/3}.$$

Proof. Consider the arc in the first quadrant. It is parametrized by

$$\gamma(t) = (1-t^2,\ 2t-t^2), \qquad 0\leq t\leq 1.$$

Indeed,

$$\sqrt{1-(1-t^2)}+\sqrt{1-(2t-t^2)} = t+(1-t) = 1.$$

Now

$$\gamma'(t) = (-2t,\ 2-2t), \qquad \gamma''(t) = (-2,\ -2),$$

hence

$$\det(\gamma'(t),\gamma''(t)) = \det\begin{pmatrix} -2t & -2 \\ 2-2t & -2\end{pmatrix} = 4.$$

Therefore the equiaffine speed is constant and equal to $4^{1/3}$. Thus one arc has equiaffine length

$$\int_0^1 4^{1/3}\,dt = 4^{1/3}.$$

Since ∂L consists of four congruent arcs, the total equiaffine length is

$$4\cdot 4^{1/3} = 4^{4/3}.$$

□

Theorem 5. *For $\Re(s)>2$,*

$$Z_L(s) = \left(8 - 2^{2-s}\frac{\zeta_{\mathrm{SU}(3)}(s)}{\zeta(3s)}\right)\frac{1}{s(s-1)}.$$

In particular, $Z_L(s)$ admits a meromorphic continuation to every point where $\zeta_{\mathrm{SU}(3)}(s)$ does.

Proof. By symmetry, the four boundary arcs of ∂L contribute the same boundary series, namely

$$F_{\partial L}(s) = 4F_{\Gamma_{\mathrm{par}}}(s).$$

Using Proposition 4, we obtain

$$F_{\partial L}(s) = 4 \cdot 2^{-s} \frac{\zeta_{\mathrm{SU}(3)}(s)}{\zeta(3s)} = 2^{2-s} \frac{\zeta_{\mathrm{SU}(3)}(s)}{\zeta(3s)}.$$

On the other hand, the minimal model of L is the square

$$\widehat{L} = [-1, 1]^2.$$

Applying the rectangle formula from the remark after Theorem 1 with

$$P = Q = 2,$$

we get

$$H_{\widehat{L}}(s) = 8.$$

Therefore Theorem 1 yields

$$s(s-1)Z_L(s) = -F_{\partial L}(s) + 8,$$

that is,

$$Z_L(s) = \left(8 - 2^{2-s} \frac{\zeta_{\mathrm{SU}(3)}(s)}{\zeta(3s)}\right) \frac{1}{s(s-1)}.$$

□

Corollary 2. *The function $Z_L(s)$ has a simple pole at $s = \frac{2}{3}$, and*

$$\operatorname*{Res}_{s=2/3} Z_L(s) = \frac{18\sqrt{3}}{\pi^3} \Gamma\left(\frac{1}{3}\right)^3.$$

Equivalently,

$$\operatorname*{Res}_{s=2/3} Z_L(s) = \frac{9\sqrt{3}}{2 \cdot 4^{1/3} \pi^3} \Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\partial L).$$

Proof. The second formula follows immediately from Proposition 5 and the general residue theorem for smooth strictly convex domains. Since

$$\mathrm{Length}_{\mathrm{equiaffine}}(\partial L) = 4^{4/3},$$

we obtain

$$\operatorname*{Res}_{s=2/3} Z_L(s) = \frac{9\sqrt{3}}{2 \cdot 4^{1/3} \pi^3} \Gamma\left(\frac{1}{3}\right)^3 \cdot 4^{4/3} = \frac{18\sqrt{3}}{\pi^3} \Gamma\left(\frac{1}{3}\right)^3.$$

□

The explicit formula above is one reason the parabola should be regarded as the basic model case. Quite remarkably, the same parabolic geometry appears in several different contexts.

5.4. **Witten's $SU(3)$ zeta function.** Following Witten and Zagier [24, 25], for a compact semisimple Lie group G, one considers the representation zeta function

$$\zeta_G(s) := \sum_{\rho \in \mathrm{Irrep}(G)} (\dim \rho)^{-s}.$$

In the case $G = SU(2)$, the irreducible representations have dimensions $1, 2, 3, \ldots$, so $\zeta_{SU(2)}(s)$ is simply the Riemann zeta function. The next case,

$$\zeta_{SU(3)}(s),$$

is exactly the one that appears in the parabolic model. Indeed, irreducible representations of $SU(3)$ are indexed by pairs $(p, q) \in \mathbb{N}^2$, and

$$\dim V_{p,q} = \frac{1}{2}\, pq(p+q).$$

Therefore

$$\zeta_{SU(3)}(s) = 2^s \sum_{p,q \geq 1} \frac{1}{\big(pq(p+q)\big)^s}.$$

Our computation shows that the boundary zeta series of the parabola is precisely the primitive version of this double series, obtained by dividing by $\zeta(3s)$.

The analytic structure of $\zeta_{SU(3)}$ was studied in [21]. In particular, $\zeta_{SU(3)}(s)$ admits a meromorphic continuation to $\mathbb{C}$, with simple poles at

$$s = \frac{2}{3}, \qquad s = \frac{3}{2} - k \quad (k \in \mathbb{N}),$$

and

$$\operatorname*{Res}_{s=2/3} \zeta_{SU(3)}(s) = \frac{4^{1/3}}{2\pi\sqrt{3}}\, \Gamma\!\left(\frac{1}{3}\right)^3.$$

This is the same transcendental factor that appears in our main residue formula, up to the elementary normalization coming from the geometric setup.

5.5. **Moduli space volume.** The same triangle-cutting mechanism also appears in the algebro-geometric story behind the universal elliptic curve over the moduli space A_1. On the one hand, for moduli spaces A_g the volume is interpreted as a self-intersection number of the Hodge bundle; on the other hand, for the universal elliptic curve $B_1 \to A_1$, Kramer and von Pippich explain that one compares the self-intersection $L_1 \cdot L_1$ of the distinguished line bundle L_1 with the Chern–Weil integral

$$\mathrm{vol}(B_1) = \int_{B_1} c_1(L_1, \|\cdot\|_\Theta) \wedge c_1(L_1, \|\cdot\|_\Theta),$$

and the defect is measured by the special value $\zeta_{\mathrm{MT}}(2, 2; 2)$ [16]. More precisely, after compactifying B_1 over the cusp and resolving the nodal boundary fiber by an infinite chain of blow-ups, each exceptional divisor contributes a correction term, so that the discrepancy is expressed by a Mordell–Tornheim series [16, 4].

What is especially close to our picture is that, in the toric model extracted in [4], self-intersection is literally computed by areas of convex regions: if $\Delta = \mathrm{conv}\{(0,0),(1,0),(0,1)\}$, then

$$\mathrm{div}(x_0)^2 = 2\,\mathrm{Vol}(\Delta) = 1,$$

whereas for the singular metric one obtains a b-divisor with

$$\text{b-}\,\mathrm{div}(x_0, \|\cdot\|_{\mathrm{sing}})^2 = 2\,\mathrm{Vol}(\Delta_{\mathrm{sing}}), \qquad \Delta_{\mathrm{sing}} = \{(x,y) \in \mathbb{R}^2_{\geq 0} :\ x+y \leq 1,\ \sqrt{x}+\sqrt{y} \geq 1\}.$$

Hence

$$\mathrm{div}(x_0)^2 - \text{b-}\,\mathrm{div}(x_0, \|\cdot\|_{\mathrm{sing}})^2 = 2\,\mathrm{Vol}(\Delta) - 2\,\mathrm{Vol}(\Delta_{\mathrm{sing}}) = 2\int_0^1 (1-\sqrt{x})^2\,dx = \frac{1}{3}.$$

Equivalently, this is the statement

$$\zeta(2,2;2) = \frac{1}{3}\,\zeta(6),$$

and geometrically the relevant boundary curve is exactly the parabola

$$\sqrt{x} + \sqrt{y} = 1.$$

This is why, in our setting, the parabola should be regarded as the model case: it is the same curve along which the infinite resolution creates the triangular defect terms whose total contribution is governed by the Mordell–Tornheim zeta value [16, 4].

5.6. **Concentration of measure.** There is also a probabilistic reason for singling out this domain. Bárány proved [2] that for convex lattice polygons in the square the overwhelming majority of them approach a deterministic limit shape, and this limit shape is exactly

$$L = \Big\{(x,y) \in \mathbb{R}^2 :\ \sqrt{1-|x|} + \sqrt{1-|y|} \geq 1\Big\}.$$

More precisely, if

$$\mathcal{P}_n = \Big\{P \subset [-1,1]^2 :\ P \text{ is a convex } (1/n)\mathbb{Z}^2\text{-lattice polygon}\Big\},$$

then for every $\varepsilon > 0$,

$$\frac{\#\{P \in \mathcal{P}_n :\ |P \triangle L| > \varepsilon\}}{\#\mathcal{P}_n} \longrightarrow 0 \qquad (n \to \infty).$$

Thus the same domain L that appears naturally in our explicit tropical-zeta computation is also the limit shape in a classical lattice-polygon problem.

Appendix A. Basic geometry of two-dimensional case

A.1. **A minimal model.** We begin with the simplest support configuration, namely the rectangular box determined by the horizontal and vertical supporting lines. The idea is that the global integral identity will be obtained by starting from this box and then inserting the remaining supporting lines one by one. Each new supporting line cuts off a single triangular region, and the next lemma computes the contribution of the initial rectangle.

Let

$$R = [0,P] \times [0,Q], \qquad P \geq Q > 0,$$

and define

$$\rho_R(x,y) := \min(x, P-x, y, Q-y).$$

Lemma 10 (Rectangle identity)**.** *For* $\Re(s) > 2$,

$$s(s-1)\int_R \rho_R(x,y)^{\,s-2}\,dx\,dy = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1}.$$

Proof. For $0 \le t \le Q/2$, the superlevel set of ρ_R is

$$\{\rho_R \ge t\} = [t, P-t] \times [t, Q-t],$$

and therefore

$$\mathrm{Area}\{\rho_R \ge t\} = (P-2t)(Q-2t).$$

By the layer-cake formula,

$$\int_R \rho_R(x,y)^{\,s-2}\,dx\,dy = (s-2)\int_0^{Q/2} t^{s-3}(P-2t)(Q-2t)\,dt.$$

Expanding the integrand gives

$$(s-2)\int_0^{Q/2} t^{s-3}\big(PQ - 2(P+Q)t + 4t^2\big)\,dt.$$

Integrating term by term and simplifying yields

$$s(s-1)\int_R \rho_R(x,y)^{\,s-2}\,dx\,dy = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1},$$

as claimed. □

We next isolate the local effect of inserting a single additional supporting line. After a unimodular affine normalization, every such local move is reduced to cutting the first quadrant by a line of the form $x + y = \lambda$. The following identity computes exactly the change of the integral under this elementary operation.

need a picture here! let it be a picture of rectangle with the tropical distance function and on the same picture we cut of or the corners and it is visible what kind of tetrahendron is cut from it

Lemma 11 (one-cut identity)**.** *Let* $\lambda > 0$, *and define*

$$\phi_-(x,y) := \min(x,y) \qquad \text{on } \{x \ge 0,\ y \ge 0\},$$

and

$$\phi_+(x,y) := \min(x,y,x+y-\lambda) \qquad \text{on } \{x \ge 0,\ y \ge 0,\ x+y \ge \lambda\}.$$

Then for $\Re(s) > 2$,

$$\int_{\{x+y<\lambda\}} \phi_-(x,y)^{\,s-2}\,dx\,dy + \int_{\{x\ge 0,\ y\ge 0,\ x+y\ge\lambda\}} \Big(\phi_-(x,y)^{\,s-2} - \phi_+(x,y)^{\,s-2}\Big)\,dx\,dy = \frac{\lambda^s}{s(s-1)}.$$

Proof. For $t \ge 0$, the difference of the superlevel sets appearing on the left-hand side is

$$E_t := \{x \ge t,\ y \ge t,\ x+y \le \lambda + t\}.$$

After the change of variables $x' = x - t$, $y' = y - t$, this becomes

$$\{x' \ge 0,\ y' \ge 0,\ x' + y' \le \lambda - t\}.$$

Hence, for $0 \le t \le \lambda$,

$$\mathrm{Area}(E_t) = \frac{(\lambda - t)^2}{2},$$

while for $t > \lambda$ the set E_t is empty.

Applying the layer-cake formula, we find that the left-hand side equals

$$(s-2)\int_0^\lambda t^{s-3}\frac{(\lambda-t)^2}{2}\,dt.$$

Substituting $t = \lambda u$ gives

$$\frac{s-2}{2}\lambda^s \int_0^1 u^{s-3}(1-u)^2\,du.$$

Since

$$\int_0^1 u^{s-3}(1-u)^2\,du = B(s-2,3) = \frac{2}{s(s-1)(s-2)},$$

we obtain

$$\frac{\lambda^s}{s(s-1)}.$$

□

A.2. **Tropical zeta functions of symplectic minimal models.** For a planar compact convex domain Ω, there are two scenarios of how the tropical wave front evolution might terminate: either with a point or with a segment. The first case corresponds to the symplectic minimal model which is m_Ω-rescaling of a reflexive polygon, i.e. a lattice polygon having only one lattice point in the interior. Up to automorphisms of the lattice, there are sixteen types of such polygons, consisting of five triangles (of lattice perimeters 3, 9, 6, 4, and 8), seven quadrilaterals (of lattice perimeters 5, 7, 6, 4 –appearing twice, and 8 – appearing twice), three pentagons (of lattice perimeters 5,7, and 6), and one hexagon (of lattice perimeter 6). This perimeter and the total time of the wave front propagation m_Ω are the only inputs we need to compute the tropical zeta function of such a minimal model.

The following proposition follows from Lemma 11.

Proposition 6. *Assume that $M_\Omega = \Omega(m_\Omega)$ consists of a single point. Then, $Z_{\hat\Omega}(s) = \frac{1}{s(s-1)}\,\mathrm{Length}_{\mathbb{Z}}(\partial\hat\Omega)m_\Omega^s$.*

Note that the above function is holomorphic outside $s = 0$ or $s = 1$, and the value at $s = 2$ indeed corresponds to the area of $\hat\Omega$.

The case of $M_\Omega = \Omega(m_\omega)$ being a segment is more involved, as now, in addition to the old continuous parameter m_Ω of the minimal model, we have the length of M_Ω that we denote by l_Ω, and three more independent integer parameters in the general case. There is also the most degenerate case, with a single integer parameter, when all the edges of the caustic of Ω have weight 2, and thus $\hat\Omega = \Omega$, and a family interpolating between the most degenerate case and the general one, which has two integer parameters labeling a type of the minimal model.

Concretely, a minimal model of degenerate type can be represented as a convex hull of points $(-l/2+mk_1, m)$, $(l/2+mk_2, m)$, $(l/2-m(k_2+1), -m)$, and $(-l/2-m(k_2+1), -m)$, where we further assume that $l \ge m|k_1-k_2|$. Here, the parameters are positive reals l, representing the length of the

central horizontal segment M_Ω, and m representing the total time of the wave front propagation, as well as integers k_1 and k_2. Note, however, that such minimal models with the same value of $|k_1 - k_2|$ are $\mathrm{GL}_2(\mathbb{Z})$-equivalent.

Proposition 7. *For the minimal model of the above paragraph, its tropical zeta function is equal to* $(s-1)^{-1}(2l + 4ms^{-1})m^{s-1}$.

The most general segment-type minimal model can be represented as the convex hull of points $(-l/2 + mn_4, m)$, $(l/2 + mn_1, m)$, $(l/2 + m, 0)$, $(l/2 + mn_2, -m)$, $(-l/2 + mn_3, -m)$, and $(-l/2 - m, 0)$, where the integer parameters n_1, n_2, n_3, n_4 satisfy some inequalities (please add them) to ensure the non-degeneracy and have a single redundancy (please find it) due to the automorphisms of the lattice.

Proposition 8. *The tropical zeta function of the above domain is equal to* $(s-1)^{-1}(2l + (4 + n_1 + n_2 - n_3 - n_4)ms^{-1})m^{s-1}$.

Finally, the mixed-type minimal model can be represented as the convex hull of $(l/2+mn_1, m)$, $(l/2+m, 0)$, $(l/2+mn_2, -m)$, $(-l/2-m(k+1), -m)$, and $(-l/2 + mk, m)$, where k, n_1, n_2 are integer parameters.

Proposition 9. *The tropical zeta function of the above minimal model is equal to* $(s-1)^{-1}(2l + (4 + n_1 + n_2)ms^{-1})m^{s-1}$. *please double check*

As a corollary of the above propositions, via the triangle cutting procedure, we have the following.

Theorem 6. *For any planar compact convex domain Ω with non-empty interior, wee have*

$$\operatorname*{Res}_{s=1} Z_\Omega(s) = \mathrm{Length}_{\mathbb{Z}}\, \partial\Omega.$$

In addition, if Ω is a unimodluar polygon corresponding to a compact smooth toric surface with canonical class K, then

$$\operatorname*{Res}_{s=0} Z_\Omega(s) = K^2.$$

Thus, for a general domain, we may think of the residue at 1 as of the symplectic area of the anticanonical class, and of the residue at 0 as of its self-intersection. Observe agian, that the former is a linear geoemtric invariant with respect to rescaling of the domain (which is the same as the rescaling of the symplectic form), and the latter is purely topological, i.e. of homogenuous degree zero, due to our normalization convention. Is the res at 0 still the same thing as the self-intersection in non-unimodluar case, say if we are within the stable range of only An singularities???

A.3. **The integral–boundary identity.** need a picture here! say the picture of cutting procedure. Start with a minimal model which is pentagon, and then do two layers of cutting of one corner (cut the corner and then cut two new corners)

We now assemble the global identity. Starting from the rectangular support box R, we insert the remaining supporting lines in Stern–Brocot order. The rectangle identity gives the initial contribution, while the one-cut identity shows that each newly inserted support line subtracts exactly one term of the boundary Dirichlet series.

Proof of theorem 1. Let $\widehat{\Omega}$ be the minimal model of Ω. By construction, Ω is obtained from $\widehat{\Omega}$ by a sequence of unimodular corner cuts in Stern–Brocot order. Let $\Omega^{(N)}$ be the polygon obtained after the first N cuts, so that

$$\widehat{\Omega} = \Omega^{(0)} \supset \Omega^{(1)} \supset \cdots \supset \Omega, \qquad \Omega^{(N)} \downarrow \Omega.$$

At each step, one inserts the mediant supporting line between two neighboring supporting lines with inward primitive normals (a, b) and (c, d), where $ad - bc = 1$. After an affine change of coordinates in $SL(2, \mathbb{Z})$, the two old supporting lines become

$$x = 0, \qquad y = 0,$$

and the new supporting line becomes

$$x + y = \lambda, \qquad \lambda = \left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|.$$

Because the change of coordinates lies in $SL(2, \mathbb{Z})$, it preserves the Euclidean area form. Moreover, the defining property of the minimal model guarantees that this planar corner cut corresponds, on the graph of the distance function, exactly to the local model of Lemma 11. Hence each cut decreases the integral by

$$\frac{1}{s(s-1)}\left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|^s.$$

Therefore, after N cuts,

$$s(s-1)Z_{\Omega^{(N)}}(s) = s(s-1)Z_{\widehat{\Omega}}(s) - \sum_{\Delta \in \mathcal{T}_N} \left(\sqrt{2\ \mathrm{Area}(\Delta)}\right)^s,$$

where $\mathcal{T}_N$ is the finite set of support triangles removed up to stage N.

As $N \to \infty$, the domains $\Omega^{(N)}$ decrease to Ω, and the corresponding distance functions $\rho_{\Omega^{(N)}}$ decrease pointwise to ρ_Ω. Since $\Re(s) > 2$, the functions $\rho_{\Omega^{(N)}}^{s-2}$ are dominated by $\rho_{\widehat{\Omega}}^{\Re(s)-2} \in L^1(\widehat{\Omega})$. Therefore, by dominated convergence,

$$Z_{\Omega^{(N)}}(s) \longrightarrow Z_\Omega(s).$$

On the other hand, for $\Re(s) > 2$, the finite sums over $\mathcal{T}_N$ converge absolutely to $F_{\partial\Omega}(s)$. Passing to the limit yields

$$s(s-1)Z_\Omega(s) = s(s-1)Z_{\widehat{\Omega}}(s) - F_{\partial\Omega}(s).$$

This is exactly

$$s(s-1)Z_\Omega(s) = -F_{\partial\Omega}(s) + H_{\widehat{\Omega}}(s),$$

with

$$H_{\widehat{\Omega}}(s) := s(s-1)Z_{\widehat{\Omega}}(s).$$

□

In particular, there exists a holomorphic function $H_\Omega(s)$, namely

$$H_\Omega(s) = 8\left(\frac{Q}{2}\right)^s + 2s(P-Q)\left(\frac{Q}{2}\right)^{s-1},$$

such that

$$s(s-1)Z_\Omega(s) = -F_{\partial\Omega}(s) + H_\Omega(s).$$

Thus, in dimension 2, the interior zeta function is determined by the boundary Dirichlet series up to an explicit holomorphic correction term coming

from the initial rectangular box. In particular, all nontrivial singular behavior of $Z_\Omega(s)$ is already encoded on the boundary.

A.4. **Exact mean-value identity.** This lemma is a standard two-point second-order mean value formula but we prove it for completeness.

Lemma 12. *Let $f \in C^2([0,1])$, let $x \neq y$ in $[0,1]$, and let $\lambda \in (0,1)$. Put*

$$m = \lambda x + (1-\lambda)y.$$

Then there exists ξ between x and y such that

$$f(m) - \lambda f(x) - (1-\lambda)f(y) = -\frac{1}{2}\lambda(1-\lambda)f''(\xi)(x-y)^2.$$

Proof. Define

$$\varphi(t) := f\big(y + t(x-y)\big), \qquad t \in [0,1].$$

Then $\varphi \in C^2([0,1])$, and

$$\varphi(\lambda) = f(m), \qquad \varphi(1) = f(x), \qquad \varphi(0) = f(y).$$

Thus it is enough to prove that there exists $\eta \in (0,1)$ such that

$$\varphi(\lambda) - \lambda\varphi(1) - (1-\lambda)\varphi(0) = -\frac{1}{2}\lambda(1-\lambda)\varphi''(\eta).$$

Set

$$L(t) := \varphi(0) + t\big(\varphi(1) - \varphi(0)\big),$$

and let

$$A := \varphi(\lambda) - L(\lambda).$$

Consider the auxiliary function

$$\psi(t) := \varphi(t) - L(t) - A\,\frac{t(t-1)}{\lambda(\lambda-1)}.$$

Then

$$\psi(0) = \psi(\lambda) = \psi(1) = 0.$$

By Rolle's theorem, there exist $u_1 \in (0,\lambda)$ and $u_2 \in (\lambda,1)$ such that

$$\psi'(u_1) = \psi'(u_2) = 0.$$

Applying Rolle's theorem once more to ψ', we obtain $\eta \in (u_1,u_2) \subset (0,1)$ such that

$$\psi''(\eta) = 0.$$

Since $L'' = 0$ and $\big(t(t-1)\big)'' = 2$, this gives

$$0 = \psi''(\eta) = \varphi''(\eta) - A\,\frac{2}{\lambda(\lambda-1)}.$$

Hence

$$A = \frac{1}{2}\,\lambda(\lambda-1)\varphi''(\eta) = -\frac{1}{2}\,\lambda(1-\lambda)\varphi''(\eta).$$

Finally,

$$\varphi''(t) = f''\big(y + t(x-y)\big)(x-y)^2.$$

Therefore, with

$$\xi := y + \eta(x-y),$$

we get

$$f(m) - \lambda f(x) - (1-\lambda) f(y) = -\frac{1}{2}\lambda(1-\lambda) f''(\xi)(x-y)^2.$$

□

Proof of Lemma 2. Apply Lemma 12 with

$$x = \frac{a}{b}, \qquad y = \frac{c}{d}, \qquad \lambda = \frac{b}{b+d}.$$

Since $ad - bc = 1$, we have

$$x - y = \frac{ad-bc}{bd} = \frac{1}{bd} \neq 0,$$

so the lemma applies. Moreover,

$$m = \lambda x + (1-\lambda) y = \frac{b}{b+d}\frac{a}{b} + \frac{d}{b+d}\frac{c}{d} = \frac{a+c}{b+d} = m(I).$$

Hence

$$c_I(f) = f(m) - \frac{b}{b+d} f\Big(\frac{a}{b}\Big) - \frac{d}{b+d} f\Big(\frac{c}{d}\Big) = -\frac{1}{2}\lambda(1-\lambda) f''(\xi_I)(x-y)^2$$

for some $\xi_I \in (c/d, a/b)$. Multiplying by $b+d$, we obtain

$$(b+d) c_I(f) = -\frac{1}{2}(b+d)\lambda(1-\lambda) f''(\xi_I)(x-y)^2.$$

Now

$$\lambda(1-\lambda) = \frac{bd}{(b+d)^2}, \qquad (x-y)^2 = \frac{(ad-bc)^2}{b^2d^2} = \frac{1}{b^2d^2}.$$

Substituting these identities gives

$$(b+d) c_I(f) = -\frac{f''(\xi_I)}{2\, bd(b+d)},$$

as claimed. □

Appendix B. Complex, symplectic and tropical geometries

Since the majority of the material in this article has a distinctly analytic flavor, we decided to include in this appendix very brief explanation of geometric terminology that we have casually utilized in several places, especially in the broader introductory Section 1.1. Thus, subsections B.1 and B.2 cover basics on topological and symplectic aspects of toric surfaces, where, although complex singularity theory in dimension two is not complicated, we have chosen to focus on the non-singular case for the sake of simplicity. B.3 provides a minimalistic overview of planar tropical wave fronts and their caustics, with an emphasis on how their structure theory is expressed in toric language. The content of B.4 is original, presenting an application of the main result for the question of finding the asymptotic of the lattice perimeter of the tropical wave front when time tends to zero.

B.1. **Topology of complex toric surfaces.** Compact smooth toric surfaces provide a tame and very well understood class of closed complex manifolds. The term "toric" comes from their defining property of being equivariant compactifications of the complex algebraic torus $(\mathbb{C}^*)^2$, where $\mathbb{C}^*$ denotes the multiplicative group of non-zero complex numbers. A toric surface is characterized by its fan represented by a collection of rational slope rays (one usually constructs a smooth toric surface by means of explicit gluings of affine plane charts dictated by the structure of the fan), which, in the compact case, is complete, i.e. every point on the plane is a sum of some vectors belonging to the rays, and, in the non-singular case, is unimodular, i.e. the primitive vectors spanning two consecutive rays form the basis of the square lattice. Some fundamental examples of complete unimodular fans (represented by sets of primitive vectors generating the rays) and corresponding surfaces are:

Example 1. *a)* $\{(-1,0),(0,-1),(1,1)\}$ *– projective plane* $\mathbb{C}P^2$*;*
b) $\{(-1,0),(0,-1),(0,1),(1,d)\}$, *–* $\mathbb{C}P^1$*- bundles (non-trial for* $d \neq 0$*) over* $\mathbb{C}P^1$*, also known as Hirzebruch surfaces.*

Let X denote a compact smooth toric surface. Then, then $(\mathbb{C}^*)^2$ is a dense open subset of X, and its complement is a union of the so-called boundary divisors $\{D_j\}$, each of which is a smooth embedding of the Riemann sphere $\mathbb{C}P^1$ to X. Moreover, the boundary divisors are in one-to-one correspondence with the rays in the fan of X. In fact, the fan can be reconstructed as follows. Take a non-zero real vector (a,b) and consider a path on $(\mathbb{C}^*)^2$ parametrized as $t \in (1,+\infty) \mapsto (t^a, t^b)$. Then, this path has a unique limit point in X for $t \to +\infty$ and this point belongs to a unique boundary divisor if and only if (a,b) belongs to a ray of the fan corresponding to the divisor. More generally, if (a,b) belongs to a connected component of complement to the union of all rays in the fan bounded by two rays, the limit point of the path will be the unique intersection point of the two boundary divisors corresponding to the two rays.

Crucially, the two boundary divisors intersect transversally at the unique point, and, as we are in the complex world, where the orientations of all manifolds are canonically prescribed and compatible with each other, it has multiplicity $+1$. These points are the fixed points of the torus action, and all other orbits, apart from the torus itself, are identified with copies of the complex one-dimensional algebraic torus $\mathbb{C}^*$, being a quotient of $(\mathbb{C}^*)^2$ by a one-dimensional stabilizer of some point of the orbit, concretely seen as the results of removing the two fixed points from each boundary divisor. Moreover, the second (co)homology of X is spanned by the classes $[D_j]$ with only two relations being $\sum v_j \otimes [D_j] = 0$, where v_j is the primitive vector in the direction of the fan corresponding to D_j. It is easy to see that the first and third cohomology groups of X vanish (either by showing that it is simply connected, or by providing a cell-decomposition with only even-dimensional cells), therefore, to determine the ring structure on the cohomology of X one is left with finding the self-intersections $[D_j]^2$. This self-intersection can be easily read from the fan, i.e. if we denote by v_{j-1} and v_{j+1} the two primitive vectors in the direction of two adjacent rays, where the indexing is done in the clockwise order, then $[D_j]^2 = \det(v_{j-1}, v_{j+1})$.

Blow-up of a point is an atomic birational transformation of complex surfaces. Abstractly, one replaces a point on the surface with the projectivization of the corresponding fiber of the tangent bundle (which results in a complex surface – the local model for this is the blow up of $\mathbb{C}^2$ at the origin coinciding with the total space of tautological line bundle over $\mathbb{C}P^1$ consisting of all pairs (p, l) where $l \in \mathbb{C}P^1$ is a line on $\mathbb{C}^2$ passing through the origin and containing $p \in \mathbb{C}^2$). In other words, one replaces the point with a Riemann sphere which, in addition, has the self-intersection equal to minus one. From the combinatorial perspective, blowing up a fixed point of the torus action on a toric surface corresponds to a unimodular subdivision of the fan. Concretely, if the fixed point is the intersection point of two boundary divisors D_j and D_{j+1} corresponding to rays spanned by primitive vectors v_j and v_{j+1}, we add a new ray spanned by the vector $v_j + v_{j+1}$. This new fan defines a new toric surface which is mapped to the old one via a "blow-down" contracting to a point the divisor corresponding to the new ray. Using the previous paragraph, we see that the self-intersection of this new divisor is indeed -1. For instance, Example 1 a) is the result of a blow down of Example 1 b) with $d = 1$, i.e. the removal of the ray spanned by $(0, 1)$.

The second cohomology of a compact smooth complex surface has a distinguished class – the canonical class, which may be defined topologically as the first Chern class of the cotangent bundle. In the toric situation, this class can be represented by the minus sum of classes of all boundary divisors, i.e. $K_X = -\sum_j [D_j]$. This is most elegantly explained by considering an explicit section of the second exterior power of the cotangent bundle, i.e. the holomorphic logarithmic two-form $d \log z_1 \wedge d \log z_2$, which extends from the algebraic torus to X as a meromorphic form with simple poles along all the boundary divisors. From the general perspective of algebraic geometry, the self-intersection of K is quite important, and for us, on the one hand, it corresponds to the residue of the tropical zeta function of a polygon dual to the fan of the toric surface under consideration, and, and on the other hand, to the time-derivate of the lattice perimeter of the tropical wave front.

B.2. **Symplectic toric surfaces.** In the previous paragraph, we started speaking about polygons dual to fans. A choice of such a polygon, up to translation, in terms of the toric surface corresponds to a numerically effective class in the real second cohomology of the surface, i.e. such a class that evaluates at each of the boundary divisors at a positive number. In fact, these numbers are precisely the lattice lengths of the sides of polygons, each side, being orthogonal to a ray of the fan, now is thought as an image of a boundary divisor. For instance, if we go back to the examples of the first paragraph in the previous subsection, the dual polygons of Example 1 a) are only standard unimodular triangles spanned by $(0, 0), (a, 0), (0, a)$, where $a > 0$, defining a ray in the one-dimensional second cohomology of the projective plane; and in Example 1 b) one has a fixed type of trapezoids for each d with two free length parameters forming a cone in the corresponding second cohomology. Such a cone is referred to a the Kähler cone. we need a reference...

Each class in the interior of the Kähler cone can be represented by a de Rham cohomology class of a symplectic form ω on the surface, i.e. a closed non-degenerate 2-form. In our situation, this form may be furthermore assumed to be invariant under the action of the compact subtorus T^2 of the algebraic torus, which turns our setup into that of an integrable system in classical Hamiltonian mechanics. Namely, the two-dimensional group of symmetries T^2, by Noether's general principle, gives rise to the two independent constants of motion. In other words, these two are combined into a T^2-invariant momentum map

$$\mu : X \to (\mathfrak{t}^2)^*,$$

where $(\mathfrak{t}^2)^* \cong \mathbb{R}^2$ is the dual of the (Abelian) Lie algebra of T^2, and

$$d\langle \mu, \xi \rangle = \iota_{V_\xi} \omega,$$

where for $\xi \in \mathfrak{t}$, the formula $d\langle \mu, \xi \rangle$ represents the exterior derivative of a function on X given by evaluating μ at ξ, and $\iota_{V_\xi}\omega$ is the one-form given by inserting in ω the vector field V_ξ on X given by the derivative at $t = 0$ of the flow produced by the action of the one-parametric subgroup $\{\exp(t\xi) : t \in \mathbb{R}\}$ of T^2.

Since the fan of the toric surface by the construction of the previous section belonged to $\mathfrak{t}^2$, its dual polygons, which by the duality assumption must be unimodular, lie on the dual plane $(\mathfrak{t}^2)^*$ and are seen as the images of momentum maps associated with invariant symplectic forms – this is the simplest non-trivial instance of the theorem of Delzant providing an equivalence of symplectic toric manifolds and unimodular polytopes, with faces being images of closures of orbits for the algebraic-torus action. We reiterate, that the lattice length of the side of such a polygon, which is the Euclidean length, normalized by the length of the primitive vector in the direction of this side, is computed as the integral of the symplectic form over the corresponding boundary divisor – its symplectic area. The area of the polygon, from this perspective, corresponds to the symplectic volume of the toric surface, which is the total integral $\frac{1}{2}\omega^2$.

Now we examine the notion of a symplectic blow-up. Unlike in the context of pure algebraic geometry or geometric topology, now blow-ups have *sizes*. Specifically, one has to consider the symplectic area of the exceptional divisor. Very concretely, a symplectic blow-up at a point p of a symplectic 4-manifold X with the form ω can be realized as follows. Recall that symplectic geometry has no local invariants, i.e. we may choose a (Darboux?) chart with coordiantes (x_1, y_1, x_2, y_2) centered at p so that $\omega = dx_1 \wedge dy_1 + dx_2 \wedge dy_2$. In this coordinates, take away the standard open four-ball of radius $\sqrt{R}$ – this is the source of our freedom, which is, however, limited – and contract its boundary S^3 to S^2 along Hopf map. The resulting four-manifold coincides with X outside of S^2, which is a manifestation of our exceptional divisor, and the symplectic form, being pointwise inherited from the original manifold, extends well to the exceptional divisor with its symplectic area being precisely R. please, check if i'm doing it right

What we need from all that is the following fact: unimodular corner cuts of momentum polygons correspond to symplectic blow-ups. Here, by the unimodular corner cut, we mean cutting-off a corner of a polygon by a

line orthogonal to $\nu_1 + \nu_2$, where ν_1 and ν_2 are the inward normal vectors of the two sides adjacent to the corner. Iterating such corner cuts is the central procedure of our paper, as we represent the tropical zeta functions by Dirichlet generating series of the sizes of such cuts. Two convince the reader, who might be not very well familiar with the above story, we suggest contemplating one example (as all the constructions above possess enough symmetries to imply the fact in full generality) of the simplest momentum map (which deviates slightly from our narrative since the $\mathbb{C}^2$ is not compact)

$$(z_1, z_2) \in \mathbb{C}^2 \mapsto (|z_1|^2, |z_2|^2) \in \mathbb{R}^2,$$

corresponding to the invariant symplectic form $z_1 \wedge \bar{z}_1 + z_2 \wedge \bar{z}_2$ maybe there are some coefficients. If we decide to cut the (only) corner at $(0,0)$, this corresponds to blowing-up the point $(0,0) \in \mathbb{C}^2$. The ball of radius R in $\mathbb{C}^2$ projects exactly to the triangle with vertices $(0,0), (R^2, 0)$, and $(0, R^2)$.

B.3. **Missing essentials on tropical optics.** Now we are ready to give a very brief overview of planar tropical optics that wasn't given coherently so far, but still was used occasionally used implicitly. First of all, recall that the tropical wave front of a compact convex domain Ω at time $t \geq 0$ is $\Omega_t = \rho_\Omega^{-1}[t, m_\Omega]$, where m_Ω is the maximum of the tropical distance series ρ_Ω. Fundamentally, Ω_t is a polygon with rational slope sides for $t \in (0, m_\Omega)$, thus we may speak about its dual fan and the corresponding toric surface X_t. In fact, for domains with corners, it may happen that this surface is singular, but the singularities are of A_n-type – i.e. they are resolved by replacing the singular point (in correspondence with a non-unimodular corner of the polygon) by a chain of n boundary divisors, each of which has self-intersection minus two. For such mildly singular surfaces, the canonical class K_t and the symplectic form ω_t still make sense, and therefore, we may write the following equation in the second de Rahm cohomology [1]:

$$\frac{d}{dt}[\omega_t] = K_t.$$

which is valid for each non-critical t.

A time is called critical, if the combinatorics of Ω_t, encoded by the dual fan, changes at it. The set of critical times has at most one accumulation point – 0. Moreover, the only kind of event happening at at a critical time is a blow-down (algebro-geometric, i.e. erasing of a ray in the fan) resulting in a smooth point of the toric surface. Let's see how the equation above can be used to derive one of the fundamental properties of wave fronts. write these statements (about the area, about the derivate of the area) are lemmas or propositions, not just a text, they deserve to be singled out The area of Ω_t can be written as $\frac{1}{2}[\omega_t]^2$, taking its derivative, we get $[\omega_t] \cdot K_t$, which is the minus the lattice perimeter of Ω_t, since K_t is the sum of classes of boundary divisors and the product with $[\omega_t]$ is their total symplectic area. If we take the derivative of the lattice perimeter, we get simply $-K_t^2$, thus having negative self-intersection of the canonical class corresponds to growing lattice perimeter of the wave front, zero – to a constant, and positive

[1] In some other writtings, there is an additional factor of 2π, but this is a matter of convention for how one measures the symplectic area. is that so?

– to a decreasing one. Note again, however, that this analysis is only valid between the critical times, where the surface, and thus its cohomology and the canonical class in it are constant.

Critical events form non-maximal (i.e. outside of $M_\Omega = \rho^{-1}(m_\Omega)$) vertices of the tropical caustic $\mathcal{K}_\Omega$ of Ω, which is the tropical analytic curve defined by the series ρ_Ω by considering all the points in Ω where it is not linear. All such vertices are trivalent and the value of ρ_Ω at the vertex is equal to the corresponding critical time. This implies that starting from a vertex on M_Ω and traversing a shortest path towards the limiting point of the caustic at $\partial\Omega$, one covers precisely m_Ω of lattice distance. Now, if we start moving again with the primitive velocity from M_Ω along some non-maximal edge of the caustic, and continue doing so in the same direction for m_Ω units of time, the set of such destinations is precisely the vertices of the minimal model $\hat{\Omega}$ of Ω. The caustic provides a canonical recipe of carving Ω from its minimal model $\hat{\Omega}$ via a sequence of corner cuts (or, symplectic blow-ups in the toric language) – each cut having a size equal to the critical time, this procedure is identical to what is described in A.3, but now we need to observe how the caustic is modified at every step.

We would like to finalize this discussion by making the following pedantic points concerning the stable non-unimodular case, i.e. when we are looking at Ω having only A_n-type corners. First of all, the caustic $\mathcal{K}_\Omega$ is traced by the vertices of Ω_t as t varies (all of them move with primitive velocity, which is a two-dimensional phenomenon), in addition, we need to include to this common trajectory the final locus M_Ω of the wave front in case it is a segment. Moreover, such segment is assigned by the standard convention the weight 2 since it is the lattice length of the difference of two gradients (which are opposite primitive vectors) of ρ_Ω to the two sides of the segment. In addition, following the same usual convention, an edge of $\mathcal{K}_\Omega$ traced by an A_n-type vertex of Ω_t is given the weight $n+1$, which ensures that the balancing condition at every vertex of the caustic is satisfied. An A_0-vertex of Ω_t is actually a unimodular vertex, i.e. it is an image under the momentum map of a non-singular point of the symplectic toric surface corresponding to Ω_t, and it traces a weight one edge of the caustic, the relative interior of which is thus tropically smooth.

Critical events of the wave front evolution, as it was mentioned, are only blow-downs, i.e. contractions of a disjoint collection of edges (in particular, it happens indeed that more than one edge of $\Omega_{t-\epsilon}$ is contracted when passing to $\Omega_{t+\epsilon}$ for $\epsilon > 0$, as we see, for instance in the example of the symmetric domain discussed in detail in section 5) resulting in unimodular vertices. However, only one of the two adjacent vertices of the contracting edge in the wave front can be non-unimodular. In the language of corner cuts of the minimal model, the event of contracting an edge adjacent to A_0 and A_n vertices can be interpreted in two different ways. One way would be that at this moment we perform a single weighted symplectic blow-up where instead of a round Darboux 4-ball we carve out an ellipsoid (seen as a non-unimodular triangle in the momentum domain picture) with ratio $1 : (n+1)$, and replace its boundary by an exceptional two-sphere. This, however, is not what we need for writing the tropical zeta function through

a Dirichlet generating series of the sizes of blow-ups. Instead, we perform $n+1$ consequential unimodular cuts, always going either clockwise or counterclockwise when choosing the next corner for the cutting, of the same size $c > 0$ – the critical time given by the value of ρ_Ω at the vertex of the caustic formed at the collapse of the edge connecting A_0 and A_n vertices in the propagation of the wave front, which gives the contribution of $(n+1)c^s$ for the corresponding vertex of the tropical caustic in Dirichlet series form $Z_{\partial\Omega}$ of the tropical zeta function of Ω.

B.4. **Asymptotics of lattice perimeter of tropical wave fronts.** In this subsection we study the asymptotic behavior of its lattice perimeter as $t \to 0^+$, under the same smoothness assumptions on $\partial\Omega$ as in the residue theorem at $s = \frac{2}{3}$.

Let $\widehat{\Omega}$ be the minimal model of Ω. For $t > 0$, let $\widehat{\Omega}^t$ be the polygon obtained from $\widehat{\Omega}$ by removing all corners what is the size of a corner? actually I do not undesratdn what happens here, the notation is huge. Provide an exmaple, better with pictures of all these omega t u things of size greater than t. In particular,

$$\widehat{\Omega}^t = \widehat{\Omega} \qquad \text{for } t \text{ sufficiently large.}$$

Lemma 13. *For $0 \le u < t$, the dual fan of $(\widehat{\Omega}^t)_u$ is constant. for each t? what is the quantifier of t?*

Since $\widehat{\Omega}^t$ is a unimodular polygon, it defines a symplectic toric surface. Let K_t be its canonical class and ω_t the corresponding symplectic form. The canonical evolution equation it should be given a statement and a link to to it or at least eqref and the lemma above imply that the class of the symplectic form corresponding to $(\widehat{\Omega}^t)_u$ for $0 \le u < t$ is

$$[\omega_t] + uK_t.$$

Hence the lattice perimeter of $(\widehat{\Omega}^t)_u$ is

$$\mathrm{Per}_{\mathbb{Z}}\big((\widehat{\Omega}^t)_u\big) = \mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^t) - u\,K_t^2.$$

We also need the compatibility relation between the truncated models and the actual wave fronts.

Lemma 14. *For every $u > 0$,*

$$(\widehat{\Omega}^u)_u = \Omega_u.$$

For a unimodular triangle Δ, define $\operatorname{size}\Delta = \sqrt{2\,\mathrm{Area}(\Delta)}$. Now let

$$A_\Omega(t) := \#\{\Delta : \ \operatorname{size}(\Delta) \ge t\},$$

where the count runs over the support triangles arising in the carving process. Then $\widehat{\Omega}^t$ is obtained from $\widehat{\Omega}$ by performing $A_\Omega(t)$ toric blow-ups, and each blow-up decreases K^2 by 1. Therefore

$$K_t^2 = K_{\widehat{\Omega}}^2 - A_\Omega(t).$$

Let

$$r_\Omega := \operatorname*{Res}_{s=2/3} Z_{\partial\Omega}(s).$$

Since $\rho_{\partial\Omega}(s)$ is the Dirichlet generating series of the support-triangle sizes, Perron's formula gives

$$A_\Omega(t) = \frac{3}{2}\, r_\Omega\, t^{-2/3} + o(t^{-2/3}) \qquad (t \to 0^+).$$

On the other hand, the lattice perimeter of Ω itself is zero, since $\partial\Omega$ has no straight edges. Thus the lattice perimeter of $\widehat{\Omega}^t$ is the tail of the sum of triangle sizes:

$$\mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^t) = -\int_0^t u\, dA_\Omega(u).$$

Integrating by parts, we obtain

$$\mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^t) = -tA_\Omega(t) + \int_0^t A_\Omega(u)\, du = 3r_\Omega\, t^{1/3} + o(t^{1/3}).$$

Using the toric evolution formula at $u = t$, we get

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^t) + tA_\Omega(t) + O(t),$$

and hence

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \frac{9}{2}\, r_\Omega\, t^{1/3} + o(t^{1/3}).$$

Since

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{9}{2} \operatorname*{Res}_{s=2/3} Z_{\partial\Omega}(s) = \frac{9}{2}\, r_\Omega,$$

this may be rewritten as follows.

Theorem 7. *Let $\Omega \subset \mathbb{R}^2$ be a compact convex tropical domain with sufficiently smooth boundary what is sufficiently smooth? we do not know? and with non-zero curvature. Then, as $t \to 0^+$,*

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \operatorname*{Res}_{s=2/3} Z_\Omega(s)\, t^{1/3} + o(t^{1/3}).$$

Equivalently,

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \frac{9}{2} \operatorname*{Res}_{s=2/3} Z_{\partial\Omega}(s)\, t^{1/3} + o(t^{1/3}).$$

Since the residue is proportional to the equiaffine perimeter of $\partial\Omega$, the leading asymptotic of the lattice perimeter of the tropical wave fronts recovers the equiaffine perimeter of Ω.

The Mellin interpretation is given by Proposition 1: the function $Z_\Omega(s)$ is the Mellin transform of the lattice perimeter profile of the tropical wave fronts Ω_t. More precisely, since

$$P_\Omega(t) := -\frac{d}{dt}\,\mathrm{Area}(\Omega_t)$$

satisfies why do we need this constant C here? this part seems detached a bit, can you rewrite it?

$$P_\Omega(t) = C_\Omega\, t^{1/3} + O(t^{1/3+\eta}) \qquad (t \to 0^+)$$

for some $\eta > 0$, then Proposition 1 gives

$$Z_\Omega(s) = \int_0^{M_\Omega} t^{s-2} P_\Omega(t)\, dt = \frac{C_\Omega}{s - \frac{2}{3}} + \text{holomorphic terms}.$$

In particular,

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = C_\Omega.$$

Equivalently,

$$\mathrm{Area}(\Omega) - \mathrm{Area}(\Omega_t) = \frac{3}{4}\, C_\Omega\, t^{4/3} + O(t^{4/3+\eta}).$$

Alternatively, we could have derived the same asymptotic law for the area of Ω_t by thinking of it as $\frac{1}{2}[\omega_t]^2$ and repeating the same analysis as for the perimeter, where in addition one would need to evaluate the tail of the series of areas of triangle cuts, which again can be done via integration by parts.

Appendix C. Analytic proof of Proposition 2

C.0.1. *Regularity hypotheses and analytic dependence on s.* For $s \in \mathbb{C}, x \in [0,1]$ define the function

$$B_s(x) := |f''(x)|^s = \exp\bigl(s \log |f''(x)|\bigr).$$

Then for each fixed x, $s \mapsto B_s(x)$ is entire. Moreover, for s in any compact set $K \subset \mathbb{C}$ we have uniform bounds

$$\sup_{s\in K} \|B_s\|_\infty \le M^{\sup_{s\in K} \Re(s)}, \qquad \sup_{s\in K} \|(B_s)'\|_\infty \le C_K < \infty, \tag{5}$$

since $(B_s)'(x) = s\,|f''(x)|^s\, \frac{f'''(x)}{f''(x)} \cdot \mathrm{sgn}(f''(x))$ and f'''/f'' is bounded by (3).

C.0.2. *A model endpoint-sampled series.*

Proof of Lemma 3. By Lemma 2, $Z_f(s) = 2^{-s} \sum_I |f''(\xi_I)|^s/(bd(b+d))^s$. Fix a small disk $D := \{s : |s - 2/3| \le \eta\}$ with $0 < \eta < 1/30$. Let $\sigma_- := \min_{s\in D} \Re(s) = 2/3 - \eta$.

For $s \in D$ and any $u, v \in [0,1]$, by the mean value theorem and (5),

$$\bigl||f''(u)|^s - |f''(v)|^s\bigr| \le \sup_{s\in D} \|(B_s)'\|_\infty\, |u - v| \le C_D\, |u - v|.$$

For a Farey interval $I = [c/d, a/b]$, we have $|\xi_I - a/b| \le |a/b - c/d| = 1/(bd)$. Hence for $s \in D$,

$$\left| \frac{|f''(\xi_I)|^s - |f''(a/b)|^s}{(bd(b+d))^s} \right| \le \frac{C_D}{(bd)\,(bd(b+d))^{\sigma_-}} = C_D\, b^{-(\sigma_-+1)} d^{-(\sigma_-+1)} (b+d)^{-\sigma_-}.$$

The right-hand side is independent of s and is summable over all $b, d \ge 1$ because at $\sigma_- = 2/3 - \eta$ the exponent on each of b, d is $\sigma_- + 1 > 1$. Therefore the series defining $Z_f(s) - Z_f^{\mathrm{end}}(s)$ converges uniformly on D by the Weierstrass M-test. Since each term is holomorphic in s, the sum is holomorphic on the interior of D. □

C.1. Auxiliary bounds for H_s and reduced-residue sampling.

Lemma 15 (Pointwise bound for H_s)**.** *For $1/2 < \sigma := \Re(s) < 1$ and $u \in (0,1]$ one has*

$$|H_s(u)| \le C_s\, u^{-\sigma},$$

with C_s depending continuously on s on compact subsets of $\{1/2 < \Re(s) < 1\}$.

Proof. Write $H_s(u) = u^{-s}(1+u)^{-s} + \sum_{k\geq 1}(k+u)^{-s}(k+1+u)^{-s}$. The first term has modulus at most $u^{-\sigma}$. For the tail, $\sum_{k\geq 1}(k+u)^{-\sigma}(k+1+u)^{-\sigma} \leq \sum_{k\geq 1} k^{-2\sigma} < \infty$. Since $u^{-\sigma} \geq 1$ on $(0,1]$, the tail is $\ll_s u^{-\sigma}$. □

Lemma 16 (Derivative bound for H_s)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Then there exists $C_K > 0$ such that for all $s \in K$ and all $u \in (0,1]$,*

$$\left|\partial_u H_s(u)\right| \leq C_K\, u^{-\Re(s)-1}.$$

Proof. Differentiate termwise (absolute convergence holds uniformly for $u \in [\delta, 1]$, and the $k = 0$ term controls the singularity as $u \downarrow 0$):

$$\partial_u H_s(u) = -s\sum_{k\geq 0}(k+u)^{-s-1}(k+1+u)^{-s} - s\sum_{k\geq 0}(k+u)^{-s}(k+1+u)^{-s-1}.$$

For $u \in (0,1]$ the $k = 0$ contributions satisfy $|(u)^{-s-1}(1+u)^{-s}| + |u^{-s}(1+u)^{-s-1}| \ll_K u^{-\Re(s)-1}$. For the tails $k \geq 1$ we bound each summand by $|s|\,(k+u)^{-\sigma-1}(k+1+u)^{-\sigma} + |s|\,(k+u)^{-\sigma}(k+1+u)^{-\sigma-1} \ll_K k^{-2\sigma-1}$, with $\sigma = \Re(s) > 1/2$, and $\sum_{k\geq 1} k^{-2\sigma-1} < \infty$ uniformly on K. Since $u^{-\sigma-1} \geq 1$ on $(0,1]$, the tail is also $\ll_K u^{-\sigma-1}$. □

Lemma 17 (Reduced residues: H_s averages to its integral)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Then for $s \in K$ and $b \geq 1$,*

$$\sum_{\substack{1\leq r\leq b\\(r,b)=1}} H_s(r/b) = \varphi(b)\int_0^1 H_s(u)\,du + O_K\big(b^{\Re(s)}\tau(b)\big),$$

where $\tau(b)$ is the divisor function and the implied constant is uniform for $s \in K$.

Proof. By inclusion–exclusion,

$$\sum_{\substack{1\leq r\leq b\\(r,b)=1}} H_s(r/b) = \sum_{d|b}\mu(d)\sum_{1\leq k\leq b/d} H_s\left(\frac{k}{b/d}\right).$$

Write $N = b/d$ and denote $S_s(N) := \sum_{k=1}^{N} H_s(k/N)$. We prove

$$S_s(N) = N\int_0^1 H_s(u)\,du + O_K(N^{\Re(s)}), \tag{6}$$

uniformly for $s \in K$.

Decompose $H_s(u) = u^{-s}(1+u)^{-s} + R_s(u)$, where $R_s(u) := \sum_{k\geq 1}(k+u)^{-s}(k+1+u)^{-s}$.

Let $\sigma_- := \inf_{s\in K}\Re(s) > \frac{1}{2}$. Then for $u \in [0,1]$,

$$|(k+u)^{-s}(k+1+u)^{-s}| \leq k^{-2\Re(s)} \leq k^{-2\sigma_-},$$

and

$$\left|\partial_u\big((k+u)^{-s}(k+1+u)^{-s}\big)\right| \ll_K k^{-2\sigma_- -1}.$$

Since $\sum_{k\geq 1} k^{-2\sigma_-} < \infty$ and $\sum_{k\geq 1} k^{-2\sigma_- -1} < \infty$, both the series for R_s and for $\partial_u R_s$ converge uniformly on $[0,1]$, uniformly in $s \in K$. Hence $R_s \in C^1([0,1])$ and $\sup_{s\in K}\|R_s'\|_\infty < \infty$.

A standard Riemann-sum estimate for C^1 functions yields

$$\sum_{j=1}^{N} R_s(j/N) = N \int_0^1 R_s(u)\,du + O_K(1).$$

It remains to treat the singular part $u^{-s}(1+u)^{-s} = u^{-s} + u^{1-s}Q_s(u)$, where $Q_s(u) := \frac{(1+u)^{-s}-1}{u}$ extends continuously to $u=0$ with $Q_s(0) = -s$. For $s \in K$, Q_s is C^1 on $[0,1]$ with norms bounded uniformly in s.

For

$$f_s(u) := u^{1-s}Q_s(u),$$

we have

$$f_s'(u) = (1-s)u^{-s}Q_s(u) + u^{1-s}Q_s'(u).$$

Since $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$, there exists $\sigma_+ < 1$ such that $\Re(s) \le \sigma_+$ for all $s \in K$. As Q_s, Q_s' are uniformly bounded on $[0,1]$ for $s \in K$, it follows that

$$|f_s'(u)| \ll_K u^{-\sigma_+} + 1,$$

hence

$$\int_0^1 |f_s'(u)|\,du \ll_K 1.$$

Therefore, by the standard Riemann-sum estimate for absolutely continuous functions,

$$\sum_{j=1}^{N} f_s(j/N) = N \int_0^1 f_s(u)\,du + O_K(1).$$

So the contribution of $u^{1-s}Q_s(u)$ to (6) is $O_K(1)$.

Finally, for u^{-s} we use a direct Riemann-sum estimate. For $s \in K$ (hence $\Re(s) > \frac{1}{2}$ and $s \neq 1$), the function $x \mapsto x^{-s}$ is C^1 on $[1,\infty)$ and

$$\sum_{j=1}^{N} j^{-s} = \int_1^N x^{-s}\,dx + O_K(1).$$

Indeed,

$$\sum_{j=1}^{N} j^{-s} - \int_1^N x^{-s}dx = \sum_{j=1}^{N} \int_j^{j+1} \left(j^{-s} - x^{-s}\right) dx + O_K(1),$$

and the sum is bounded in modulus by $\sum_{j\ge1} \int_j^{j+1} |(x^{-s})'|\,dx\,dt \ll_K \int_1^\infty x^{-\Re(s)-1}dx < \infty$. Multiplying by N^s yields

$$\sum_{j=1}^{N} (j/N)^{-s} = N^s \int_1^N x^{-s}dx + O_K(N^{\Re(s)}).$$

But $N^s \int_1^N x^{-s}dx = \frac{N-N^s}{1-s}$, so

$$\sum_{j=1}^{N} (j/N)^{-s} - N\int_0^1 u^{-s}du = -\frac{N^s}{1-s} + O_K(N^{\Re(s)}) = O_K(N^{\Re(s)}),$$

since $|N^s| = N^{\Re(s)}$. Combining the three pieces proves (6).

Returning to inclusion–exclusion,

$$\sum_{d|b} \mu(d) S_s(b/d) = \left(\sum_{d|b} \mu(d) \frac{b}{d} \right) \int_0^1 H_s + O_K \left(\sum_{d|b} (b/d)^{\Re(s)} \right).$$

The main coefficient equals $\varphi(b)$. The error satisfies $\sum_{d|b} (b/d)^{\Re(s)} \le b^{\Re(s)} \tau(b)$. □

C.2. **Incomplete Kloosterman sums (Weil bound).** For integers n and $b \ge 1$ define the incomplete Kloosterman sum

$$K_b(n; R) := \sum_{\substack{1 \le r \le R \\ (r,b)=1}} e\left(\frac{n\overline{r}}{b} \right), \qquad e(x) := e^{2\pi i x}.$$

Lemma 18 (Completion bound for $K_b(n; R)$)**.** *For all integers n, all $b \ge 1$, and all $1 \le R \le b$,*

$$K_b(n; R) := \sum_{\substack{1 \le r \le R \\ (r,b)=1}} e\left(\frac{n\overline{r}}{b} \right) \ll (n, b)^{1/2}\, b^{1/2}\, \tau(b)^2 \log(2b),$$

with an absolute implied constant.

In particular, if $(n, b) = 1$, then for every $\varepsilon > 0$,

$$\max_{1 \le R \le b} |K_b(n; R)| \ll_\varepsilon b^{1/2+\varepsilon}.$$

Proof. For $x \in \mathbb{Z}/b\mathbb{Z}$, define

$$w_R(x) := \sum_{m=1}^{R} \mathbf{1}_{x \equiv m \pmod b}.$$

Its discrete Fourier expansion is

$$w_R(r) = \frac{1}{b} \sum_{h \bmod b} \widehat{w}_R(h)\, e\left(\frac{hr}{b} \right), \qquad \widehat{w}_R(h) := \sum_{m=1}^{R} e\left(-\frac{hm}{b} \right).$$

Hence

$$K_b(n; R) = \sum_{\substack{r \bmod b \\ (r,b)=1}} w_R(r) e\left(\frac{n\overline{r}}{b} \right) = \frac{1}{b} \sum_{h \bmod b} \widehat{w}_R(h)\, S(n, h; b),$$

where

$$S(n, h; b) := \sum_{\substack{r \bmod b \\ (r,b)=1}} e\left(\frac{n\overline{r} + hr}{b} \right)$$

is the classical Kloosterman sum.

We separate the zero frequency $h = 0$. Since $\widehat{w}_R(0) = R$, we get

$$\frac{1}{b}\, \widehat{w}_R(0)\, S(n, 0; b) = \frac{R}{b}\, c_b(n),$$

where $c_b(n) = S(n, 0; b)$ is the Ramanujan sum.

Let $g = (n, b)$. Since

$$c_b(n) = \mu\left(\frac{b}{g}\right)\frac{\varphi(b)}{\varphi(b/g)},$$

we have

$$|c_b(n)| \le \frac{\varphi(b)}{\varphi(b/g)} \le g = (n, b).$$

Hence

$$\left|\frac{R}{b}c_b(n)\right| \le (n, b),$$

and then it follows that

$$\left|\frac{1}{b}\,\widehat{w}_R(0)\,S(n, 0; b)\right| \le (n, b).$$

Since $(n, b) \le (n, b)^{1/2}b^{1/2}$, this term is acceptable.

Now consider the nonzero frequencies $h \neq 0$. For representatives

$$-\frac{b}{2} \le h \le \frac{b}{2}, \qquad h \neq 0,$$

the geometric-series estimate gives

$$|\widehat{w}_R(h)| \ll \min\left(R, \frac{b}{|h|}\right).$$

Also, by the Weil bound for Kloosterman sums,

$$|S(n, h; b)| \ll \tau(b)\,(n, h, b)^{1/2}\,b^{1/2}.$$

Group the nonzero frequencies according to $d = (h, b)$. Then

$$(n, h, b) = (n, d).$$

Therefore

$$\sum_{\substack{h \bmod b\\ h\neq 0}} |\widehat{w}_R(h)|\,|S(n, h; b)| \ll \tau(b)b^{1/2}\sum_{d|b}(n, d)^{1/2}\sum_{\substack{h \bmod b\\ h\neq 0\\ (h,b)=d}} |\widehat{w}_R(h)|.$$

So it remains to bound the inner sum. Write $h = dh_1$. Then h_1 runs through a subset of nonzero residue classes modulo b/d, and hence

$$\sum_{\substack{h \bmod b\\ h\neq 0\\ (h,b)=d}} |\widehat{w}_R(h)| \le \sum_{1\le|m|\le b/(2d)} \min\left(R, \frac{b}{d|m|}\right) \ll \frac{b}{d}\log(2b).$$

Thus

$$\sum_{\substack{h \bmod b\\ h\neq 0}} |\widehat{w}_R(h)|\,|S(n, h; b)| \ll \tau(b)b^{1/2}\log(2b)\sum_{d|b}(n, d)^{1/2}\frac{b}{d}.$$

Dividing by b, we obtain

$$\frac{1}{b}\sum_{\substack{h \bmod b\\ h\neq 0}} |\widehat{w}_R(h)|\,|S(n, h; b)| \ll \tau(b)b^{1/2}\log(2b)\sum_{d|b}\frac{(n, d)^{1/2}}{d}.$$

Now

$$(n, d)^{1/2} \le (n, b)^{1/2},$$

so

$$\sum_{d|b} \frac{(n,d)^{1/2}}{d} \le (n,b)^{1/2} \sum_{d|b} \frac{1}{d} \le (n,b)^{1/2}\tau(b).$$

Hence the contribution of the nonzero frequencies is

$$\ll (n,b)^{1/2} b^{1/2} \tau(b)^2 \log(2b).$$

Combining the zero and nonzero frequencies gives

$$K_b(n;R) \ll (n,b)^{1/2} b^{1/2} \tau(b)^2 \log(2b),$$

as claimed.

Finally, if $(n,b)=1$, then

$$K_b(n;R) \ll b^{1/2}\tau(b)^2 \log(2b) \ll_\varepsilon b^{1/2+\varepsilon},$$

since $\tau(b)^2 \log(2b) \ll_\varepsilon b^\varepsilon$. □

C.3. **A Fejér approximation bound for periodic Lipschitz functions.** Let F_N denote the Fejér kernel on the circle,

$$F_N(t) := \frac{1}{N}\left(\frac{\sin(\pi N t)}{\sin(\pi t)}\right)^2, \qquad t \in [-1/2, 1/2],$$

extended 1-periodically.

Lemma 19 (First moment of the Fejér kernel)**.** *There exists an absolute constant $C > 0$ such that for all $N \ge 2$,*

$$\int_{-1/2}^{1/2} |t|\, F_N(t)\, dt \le C\, \frac{\log N}{N}.$$

Consequently, if G is 1-periodic and Lipschitz, then its Fejér mean $G * F_N$ satisfies

$$\|G - G * F_N\|_\infty \le C\, \frac{\log N}{N}\, \mathrm{Lip}(G).$$

Proof. Since $F_N \ge 0$ and $\int_{-1/2}^{1/2} F_N = 1$,

$$G(x) - (G * F_N)(x) = \int_{-1/2}^{1/2} \big(G(x) - G(x-t)\big) F_N(t)\, dt.$$

By Lipschitz continuity,

$$|G(x) - G(x-t)| \le \mathrm{Lip}(G)\, |t|,$$

which gives the second inequality once the moment bound is proved.

For the moment bound, split the integral into $|t| \le 1/N$ and $1/N < |t| \le 1/2$. For $|t| \le 1/N$, use $F_N(t) \le N$ to get

$$\int_{|t|\le 1/N} |t| F_N(t)\, dt \ll N \int_0^{1/N} t\, dt \ll \frac{1}{N}.$$

For $1/N < |t| \le 1/2$, use $|\sin(\pi t)| \gg |t|$ and $|\sin(\pi N t)| \le 1$ to obtain

$$F_N(t) \ll \frac{1}{N t^2}.$$

Hence

$$\int_{1/N<|t|\le 1/2} |t|F_N(t)\,dt \ll \frac{1}{N}\int_{1/N}^{1/2}\frac{dt}{t} \ll \frac{\log N}{N}.$$

□

C.4. **Proof of Proposition 2.**

Proof of Proposition 2. Fix $\varepsilon > 0$, and write $\sigma := \Re(s)$. Let

$$B_s(x) := |f''(x)|^s, \qquad A_r := H_s(r/b).$$

We may assume $b \ge 4$, since for bounded b the claim is absorbed into the implied constant. Set

$$N := R := \lfloor b^{1/2} \rfloor, \qquad \eta := \varepsilon/4.$$

Define the affine endpoint interpolant

$$\ell_s(x) := (1-x)B_s(0) + xB_s(1),$$

and the remainder

$$G_s(x) := B_s(x) - \ell_s(x).$$

Then

$$G_s(0) = G_s(1) = 0.$$

By (5), the family B_s is C^1 with norms bounded uniformly for $s \in K$. Hence

$$|B_s(1) - B_s(0)| \ll_{f,K} 1, \qquad \mathrm{Lip}(G_s) \ll_{f,K} 1,$$

uniformly for $s \in K$. In particular, G_s admits a 1-periodic Lipschitz extension to the circle.

Step 1: replace the periodic remainder G_s by its Fejér mean. By Lemma 19,

$$\|G_s - (G_s)_N\|_\infty \ll_{f,K} \frac{\log(2N)}{N},$$

uniformly for $s \in K$. Therefore

$$\Sigma_b(s) = \sum_{(r,b)=1} A_r\,(G_s)_N(\overline{r}/b) + \sum_{(r,b)=1} A_r\,\ell_s(\overline{r}/b) + O_{f,K}\left(\frac{\log(2N)}{N}\sum_{(r,b)=1}|A_r|\right).$$

Now Lemma 15 yields

$$\sum_{(r,b)=1}|A_r| \le \sum_{r=1}^{b}|H_s(r/b)| \ll_K \sum_{r=1}^{b}(r/b)^{-\sigma} = b^\sigma\sum_{r=1}^{b} r^{-\sigma} \ll_K b,$$

uniformly for $s \in K$, since $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Thus

$$\Sigma_b(s) = \sum_{(r,b)=1} A_r\,(G_s)_N(\overline{r}/b) + \sum_{(r,b)=1} A_r\,\ell_s(\overline{r}/b) + O_{f,K}\big(b^{1/2}\log(2b)\big). \tag{7}$$

Step 2: expand the periodic part in Fourier modes. Write

$$(G_s)_N(x) = \widehat{G}_s(0) + \sum_{0<|n|<N}\widehat{(G_s)_N}(n)e(nx), \qquad \widehat{G}_s(0) = \int_0^1 G_s(x)\,dx.$$

Since G_s is Lipschitz with $\mathrm{Lip}(G_s) \ll_{f,K} 1$, its Fourier coefficients satisfy

$$|\widehat{G}_s(n)| \ll_{f,K} \frac{1}{|n|} \qquad (n \ne 0),$$

uniformly for $s \in K$, and hence

$$|\widehat{(G_s)_N}(n)| \le |\widehat{G}_s(n)| \ll_{f,K} \frac{1}{|n|} \qquad (0 < |n| < N).$$

Therefore

$$\sum_{(r,b)=1} A_r\,(G_s)_N(\overline{r}/b) = \widehat{G}_s(0) \sum_{(r,b)=1} A_r + \sum_{0<|n|<N} \widehat{(G_s)_N}(n)\, S_b(n),$$

where

$$S_b(n) := \sum_{(r,b)=1} A_r\, e\!\left(\frac{n\overline{r}}{b}\right).$$

Step 3: expand the affine part exactly on the residue grid. Let

$$\ell_s(x) = \alpha_s + \beta_s x, \qquad \beta_s = B_s(1) - B_s(0).$$

For $m = 0, \dots, b-1$, the values $\ell_s(m/b)$ admit the discrete Fourier expansion

$$\ell_s(m/b) = c_{b,s}(0) + \sum_{0<|n|\le b/2} c_{b,s}(n) e(nm/b),$$

where the sum runs over any fixed symmetric system of representatives of the nonzero residue classes modulo b. The coefficients satisfy

$$c_{b,s}(0) = \frac{1}{b}\sum_{m=0}^{b-1} \ell_s(m/b) = \int_0^1 \ell_s(x)\,dx + O_{f,K}(b^{-1}),$$

and, for $0 < |n| \le b/2$,

$$c_{b,s}(n) = -\frac{\beta_s}{b\big(1 - e(-n/b)\big)}, \qquad |c_{b,s}(n)| \ll_{f,K} \frac{1}{|n|},$$

uniformly for $s \in K$. Hence

$$\sum_{(r,b)=1} A_r\, \ell_s(\overline{r}/b) = c_{b,s}(0) \sum_{(r,b)=1} A_r + \sum_{0<|n|\le b/2} c_{b,s}(n)\, S_b(n).$$

Step 4: the main term. By Lemma 17,

$$\sum_{(r,b)=1} A_r = \varphi(b) \int_0^1 H_s(u)\,du + O_K\big(b^\sigma \tau(b)\big),$$

uniformly for $s \in K$. Also,

$$\widehat{G}_s(0) + c_{b,s}(0) = \int_0^1 G_s(x)\,dx + \int_0^1 \ell_s(x)\,dx + O_{f,K}(b^{-1}) = \int_0^1 B_s(x)\,dx + O_{f,K}(b^{-1}).$$

Therefore the contribution of the zero Fourier mode is

$$\varphi(b)\left(\int_0^1 H_s(u)\,du\right)\left(\int_0^1 B_s(x)\,dx\right) + O_{f,K}\big(b^\sigma\tau(b)\big) + O_{f,K}(1).$$

Since $\sigma < 1$, the error $O_{f,K}(b^\sigma \tau(b))$ is dominated by the final error term.

Step 5: bound the oscillatory sums $S_b(n)$. Let $n \ne 0$. Split

$$S_b(n) = S_b^{\le R}(n) + S_b^{>R}(n),$$

where

$$S_b^{\le R}(n) := \sum_{\substack{1\le r\le R\\(r,b)=1}} A_r\, e\!\left(\frac{n\overline{r}}{b}\right), \qquad S_b^{>R}(n) := \sum_{\substack{R<r\le b\\(r,b)=1}} A_r\, e\!\left(\frac{n\overline{r}}{b}\right).$$

For the initial segment, Lemma 15 gives

$$|S_b^{\le R}(n)| \le \sum_{r\le R} |A_r| \ll_K b^\sigma \sum_{r\le R} r^{-\sigma} \ll_K b^\sigma R^{1-\sigma} \ll_K b^{\frac{1+\sigma}{2}}.$$

For the tail, define

$$a_r := \mathbf{1}_{(r,b)=1}\, e\!\left(\frac{n\overline{r}}{b}\right), \qquad T_n(t) := \sum_{1\le r\le t} a_r.$$

Then Lemma 18 gives

$$\max_{1\le t\le b} |T_n(t)| \ll_\eta (n,b)^{1/2} b^{1/2+\eta}.$$

By Abel summation,

$$S_b^{>R}(n) = H_s(1)\, T_n(b) - H_s(R/b)\, T_n(R) - \sum_{t=R}^{b-1} T_n(t)\Big(H_s\!\left(\frac{t+1}{b}\right) - H_s\!\left(\frac{t}{b}\right)\Big).$$

Using Lemma 15,

$$|H_s(1)| \ll_K 1, \qquad |H_s(R/b)| \ll_K (R/b)^{-\sigma}.$$

Also, by the mean value theorem and Lemma 16,

$$\left|H_s\!\left(\frac{t+1}{b}\right) - H_s\!\left(\frac{t}{b}\right)\right| \le \frac{1}{b} \sup_{u\in[t/b,(t+1)/b]} |\partial_u H_s(u)| \ll_K \frac{1}{b}\left(\frac{t}{b}\right)^{-\sigma-1} \ll_K b^\sigma t^{-\sigma-1}.$$

Hence

$$\sum_{t=R}^{b-1} \left|H_s\!\left(\frac{t+1}{b}\right) - H_s\!\left(\frac{t}{b}\right)\right| \ll_K b^\sigma \sum_{t\ge R} t^{-\sigma-1} \ll_K b^\sigma R^{-\sigma}.$$

Therefore

$$|S_b^{>R}(n)| \ll_{K,\eta} (n,b)^{1/2} b^{1/2+\eta}\Big(1 + (R/b)^{-\sigma} + b^\sigma R^{-\sigma}\Big) \ll_{K,\eta} (n,b)^{1/2} b^{\frac{1+\sigma}{2}+\eta},$$

since $R \asymp b^{1/2}$. Together with the initial-segment bound, this gives

$$S_b(n) \ll_{K,\eta} (n,b)^{1/2} b^{\frac{1+\sigma}{2}+\eta} \qquad (n \ne 0). \tag{8}$$

Step 6: sum the nonzero Fourier modes. For the periodic part, using (8),

$$\sum_{0<|n|<N} |\widehat{(G_s)_N}(n)\, S_b(n)| \ll_{f,K,\eta} b^{\frac{1+\sigma}{2}+\eta} \sum_{1\le n<N} \frac{(n,b)^{1/2}}{n}.$$

Grouping by $d = (n,b)$, we obtain

$$\sum_{1\le n<N} \frac{(n,b)^{1/2}}{n} \le \sum_{d|b} d^{-1/2} \sum_{m<N/d} \frac{1}{m} \ll \tau(b)\log(2N).$$

Hence

$$\sum_{0<|n|<N} \widehat{(G_s)_N}(n)\, S_b(n) = O_{f,K,\eta}\Big(b^{\frac{1+\sigma}{2}+\eta}\tau(b)\log(2N)\Big).$$

Similarly, for the affine part,

$$\sum_{0<|n|\le b/2} |c_{b,s}(n)\, S_b(n)| \ll_{f,K,\eta} b^{\frac{1+\sigma}{2}+\eta} \sum_{1\le n\le b/2} \frac{(n,b)^{1/2}}{n} \ll b^{\frac{1+\sigma}{2}+\eta}\tau(b)\log(2b).$$

Since $\tau(b)\log(2b) \ll_\eta b^{2\eta}$, both nonzero-mode contributions are

$$O_{f,K,\varepsilon}\Big(b^{\frac{1+\sigma}{2}+\varepsilon}\Big).$$

The truncation error in (7) also satisfies

$$b^{1/2}\log(2b) \ll_{f,K,\varepsilon} b^{\frac{1+\sigma}{2}+\varepsilon},$$

because $\sigma > \frac{1}{2}$.

Putting everything together, and recalling that

$$\int_0^1 B_s(x)\,dx = \int_0^1 |f''(x)|^s\,dx,$$

we conclude that

$$\Sigma_b(s) = \varphi(b)\left(\int_0^1 H_s(u)\,du\right)\left(\int_0^1 |f''(x)|^s\,dx\right) + O_{f,K,\varepsilon}\Big(b^{\frac{1+\Re(s)}{2}+\varepsilon}\Big),$$

uniformly for $s \in K$. This proves the proposition. □

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